



A characterization of graphs with maximum cycle isolation number

Siyue Chen, Qing Cui^{*}, Jingshu Zhang

School of Mathematics, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, PR China

ARTICLE INFO

Article history:

Received 3 June 2024

Received in revised form 11 October 2024

Accepted 21 January 2025

Available online 28 January 2025

Keywords:

Partial domination

Isolating set

Isolation number

Extremal graphs

ABSTRACT

For any graph G , a subset D of vertices of G is called a cycle isolating set of G if $G - N[D]$ contains no cycle. The cycle isolation number of G , denoted by $\iota_c(G)$, is the minimum cardinality of a cycle isolating set of G . Analogously, for any positive integer k and any graph G , a subset D of vertices of G is called a K_k -isolating set of G if $G - N[D]$ does not contain K_k as a subgraph. The K_k -isolation number of G , denoted by $\iota(G, K_k)$, is the minimum cardinality of a K_k -isolating set of G . Borg (2020) proved that if G is a connected n -vertex graph and $G \not\cong C_3$, then $\iota_c(G) \leq \frac{n}{4}$. Borg, Fenech and Kaemawichanurat (2020) showed that if G is a connected n -vertex graph, then $\iota(G, K_k) \leq \frac{n}{k+1}$ unless $G \cong K_k$ or $k = 2$ and $G \cong C_5$. These settled two problems of Caro and Hansberg (2017), and both bounds are sharp. Borg, Fenech and Kaemawichanurat (2020) further asked for a characterization of all connected n -vertex graphs G such that $\iota(G, K_k) = \frac{n}{k+1}$. The case when $k = 2$ for this problem was recently solved by Boyer and Goddard (2024). In this paper, we characterize all connected n -vertex graphs G such that $\iota_c(G) = \frac{n}{4}$. As a consequence, we also characterize all connected n -vertex graphs G such that $\iota(G, K_3) = \frac{n}{4}$. This solves the aforementioned problem of Borg, Fenech and Kaemawichanurat (2020) for the case $k = 3$.

© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

In this paper we only consider finite simple graphs. For a graph G , we use $V(G)$ and $E(G)$ to denote the vertex set and the edge set of G , respectively. For any $v \in V(G)$, the *open neighborhood* $N_G(v)$ of v is the set of neighbors of v in G , and the *closed neighborhood* $N_G[v]$ of v is the set $N_G(v) \cup \{v\}$. The *degree* of a vertex $v \in V(G)$ is $d_G(v) = |N_G(v)|$. The minimum degree and the maximum degree of G are denoted by $\delta(G)$ and $\Delta(G)$, respectively. The *open neighborhood* of a set $S \subseteq V(G)$ is the set $N_G(S) = \bigcup_{v \in S} N_G(v) \setminus S$, and the *closed neighborhood* of S is the set $N_G[S] = N_G(S) \cup S$. For any $S \subseteq V(G)$, we denote by $G[S]$ and $G - S$ the subgraphs of G induced by S and $V(G) \setminus S$, respectively, and write $G - s$ instead of $G - \{s\}$ if $S = \{s\}$. For any $A, B \subseteq V(G)$ with $A \cap B = \emptyset$, let $E_G(A, B)$ denote the set of edges of G with one endvertex in A and the other endvertex in B . We may omit the subscript G if it is clear from the context. For two graphs G and H and a positive integer t , let $G \cup H$ denote the disjoint union of G and H , and tG denote the disjoint union of t copies of G . We use P_n , C_n , K_n and S_n to denote a path, a cycle, a complete graph and a star with n vertices, respectively. Let C_n^+ be the graph obtained by attaching a pendant edge to one vertex of C_n , and K_n^- the graph obtained from K_n by removing one edge. For any positive integer k , we denote by $[k]$ the set $\{1, 2, \dots, k\}$. We write $A := B$ to rename B as A .

Let G be a graph and $\mathcal{F}$ be a set of graphs. A subset D of vertices of G is called an $\mathcal{F}$ -isolating set of G if $G - N[D]$ does not contain any member of $\mathcal{F}$ as a subgraph. The $\mathcal{F}$ -isolation number of G , denoted by $\iota(G, \mathcal{F})$, is the minimum cardinality of

^{*} Corresponding author.

E-mail addresses: chensiyue@nuaa.edu.cn (S. Chen), cui@nuaa.edu.cn (Q. Cui), zhangjs@nuaa.edu.cn (J. Zhang).

an $\mathcal{F}$ -isolating set of G . If $\mathcal{F} = \{F\}$, then an $\mathcal{F}$ -isolating set of G is simply called an F -isolating set of G and the $\mathcal{F}$ -isolation number of G is simply called the F -isolation number of G , and moreover, we simply write $\iota(G, F)$ instead of $\iota(G, \mathcal{F})$.

The study of $\mathcal{F}$ -isolating sets of graphs was first introduced by Caro and Hansberg [12]. It can be viewed as a natural generalization of the classical domination problems [17–24]. Recall that a subset D of vertices of a graph G is called a *dominating set* of G if every vertex in $V(G) \setminus D$ is adjacent to at least one vertex in D . The *domination number* of G , denoted by $\gamma(G)$, is the minimum cardinality of a dominating set of G . It is clear that D is a dominating set of G if and only if D is a K_1 -isolating set of G , and thus $\gamma(G) = \iota(G, K_1)$.

A classical result of Ore [29] asserts that the domination number of every connected n -vertex graph with $n \geq 2$ is at most $\frac{n}{2}$. Then, it follows from $\gamma(G) = \iota(G, K_1)$ that $\iota(G, K_1) \leq \frac{n}{2}$ for every connected n -vertex graph G with $G \not\cong K_1$. Caro and Hansberg [12] further showed that if G is a connected n -vertex graph and $G \notin \{K_2, C_5\}$, then $\iota(G, K_2) \leq \frac{n}{3}$. These were later generalized by Borg, Fenech and Kaemawichanurat [6] for any positive integer k .

Theorem 1.1. (Borg, Fenech and Kaemawichanurat [6]) *If G is a connected n -vertex graph, then $\iota(G, K_k) \leq \frac{n}{k+1}$ unless $G \cong K_k$ or $k = 2$ and $G \cong C_5$. Moreover, this bound is sharp.*

Theorem 1.1 settled a problem of Caro and Hansberg [12]. (Another sharp upper bound on $\iota(G, K_k)$ in terms of the number of edges was also given by Borg, Fenech and Kaemawichanurat in [7].) At the end of their paper, Borg, Fenech and Kaemawichanurat [6] asked for a characterization of all extremal graphs attaining the bound stated in Theorem 1.1.

Problem 1.2. (Borg, Fenech and Kaemawichanurat [6]) *Characterize all connected n -vertex graphs G such that $\iota(G, K_k) = \frac{n}{k+1}$.*

Payan and Xuong [30], and independently Fink et al. [16] determined all connected n -vertex graphs G such that $\gamma(G) = \frac{n}{2}$. This answers Problem 1.2 for the case $k = 1$. Building on work of Lemańska, Mora and Souto-Salorio [28], the case $k = 2$ was recently settled by Boyer and Goddard [10]. In this paper, we shall solve this problem for the case $k = 3$ (see Theorem 1.5).

For this purpose, we deal with another type of isolation of graphs. For any graph G , a subset D of vertices of G is called a *cycle isolating set* of G if $G - N[D]$ contains no cycle. The *cycle isolation number* of G , denoted by $\iota_c(G)$, is the minimum cardinality of a cycle isolating set of G . Thus, $\iota_c(G) = \iota(G, \mathcal{C})$, where $\mathcal{C} = \{C_k : k \geq 3\}$. Settling another problem of Caro and Hansberg [12], Borg [2] proved the following result.

Theorem 1.3. (Borg [2]) *If G is a connected n -vertex graph and $G \not\cong C_3$, then $\iota_c(G) \leq \frac{n}{4}$. Moreover, this bound is sharp.*

It is worth mentioning that Theorem 1.3 was later proved in stronger forms by Borg [3,4] and Zhang and Wu [37]. In [41], Zhang and Wu further proved that $\iota_c(G) \leq \frac{n}{5}$ for every connected triangle-free n -vertex graph G with $G \not\cong C_4$. Sharp upper bounds on $\iota_c(G)$ in terms of the number of edges were given in [5,14,38,40]. In [4], Borg also suggested to investigate the C_k -isolation of graphs. It follows from Theorem 1.1 that $\iota(G, C_3) = \iota(G, K_3) \leq \frac{n}{4}$ for every connected n -vertex graph G with $G \not\cong C_3$. Very recently, Bartolo, Borg and Scicluna [1] proved that $\iota(G, C_4) \leq \frac{n}{5}$ for every connected n -vertex graph G , apart from nine exceptional graphs (including C_4 , K_4^- , K_4 and six 9-vertex graphs). For $k \geq 5$, a sharp upper bound on $\iota(G, C_k)$ is still unknown.

There also have been some interesting results on other types of isolation of graphs. Zhang and Wu [36,37] proved that $\iota(G, P_3) \leq \frac{2n}{7}$ for every connected n -vertex graph G with $G \notin \{P_3, C_3, C_6\}$, and $\iota(G, P_4) \leq \frac{n}{4}$ for every connected n -vertex graph G with $G \not\cong C_7$. Chen and Xu [13] showed that if G is a connected n -vertex graph and $G \not\cong C_8$, then $\iota(G, P_5) \leq \frac{2n}{9}$. Yan [33] proved that $\iota(G, K_4^-) \leq \frac{n}{5}$ for every connected n -vertex graph G , apart from three exceptional graphs (including K_4^- , K_4 and a 9-vertex graph). More results can be found in [8,9,11,15,25–27,31,32,34,35,39].

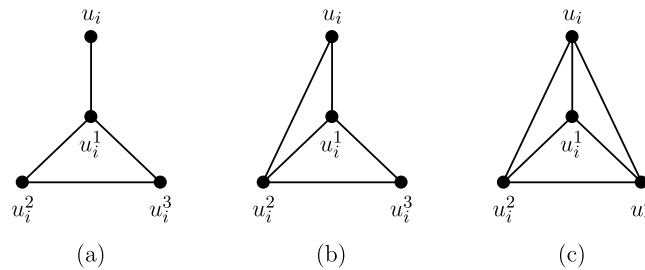
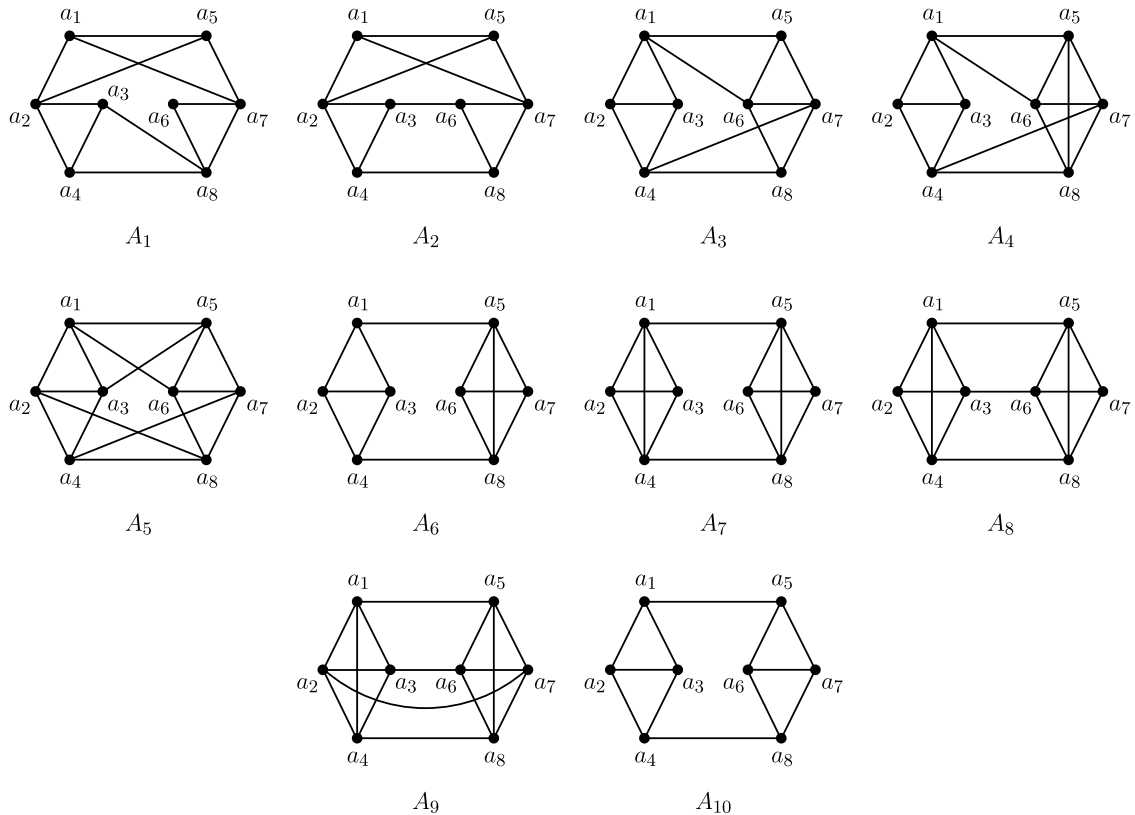
In this paper, we characterize all extremal graphs attaining the bound stated in Theorem 1.3. As a consequence, we also characterize all connected n -vertex graphs G such that $\iota(G, K_3) = \frac{n}{4}$ (This solves Problem 1.2 for the case $k = 3$). In order to state our main results, we need to introduce two sets of graphs.

For any positive integer t , let F be an arbitrary connected t -vertex graph with $V(F) = \{u_i : 1 \leq i \leq t\}$, and let $H_1, H_2, \dots, H_t$ be t vertex-disjoint copies of C_3 such that $H_i := u_i^1 u_i^2 u_i^3$ for each $i \in [t]$. Let G be the graph obtained from F and $H_1, H_2, \dots, H_t$, in which u_i is joined to H_i (for each $i \in [t]$) with one of the following three ways:

- (i) u_i is only adjacent to u_i^1 (see Fig. 1(a));
- (ii) u_i is adjacent to both u_i^1 and u_i^2 (see Fig. 1(b));
- (iii) u_i is adjacent to u_i^1, u_i^2 and u_i^3 (see Fig. 1(c)).

For each $i \in [t]$, we say that $G_i := G[V(H_i) \cup \{u_i\}]$ is a C_3 -constituent of G and u_i is the C_3 -connection of G_i in G . It is easy to observe that $G_i \in \{C_3^+, K_4^-, K_4\}$ for each $i \in [t]$ and $\{u_i^1 : 1 \leq i \leq t\}$ is a dominating set of G .

Let $\mathcal{G}_2$ be the set of all graphs G which can be constructed from the above ways, and let $\mathcal{G}_1$ be the set of graphs depicted in Fig. 2. Our main results are as follows.

Fig. 1. The three ways of joining u_i to H_i .Fig. 2. The graphs in $\mathcal{G}_1$.

Theorem 1.4. For any connected n -vertex graph G , $\iota_c(G) = \frac{n}{4}$ if and only if $G \in \{C_4\} \cup \mathcal{G}_1 \cup \mathcal{G}_2$.

Theorem 1.5. For any connected n -vertex graph G , $\iota(G, K_3) = \frac{n}{4}$ if and only if $G \in (\mathcal{G}_1 \setminus \{A_2\}) \cup \mathcal{G}_2$.

The rest of the paper is organized as follows. In Section 2, we state several preliminary results and introduce some structural properties of the graphs in $\mathcal{G}_1$. The proofs of Theorems 1.4 and 1.5 will be given in Sections 3 and 4, respectively.

2. Preliminaries

In this section, we state several preliminary results which will be used in the next section to prove Theorem 1.4. The first lemma was proved by Zhang and Wu in [40].

Lemma 2.1. (Zhang and Wu [40]) Let G be a graph and S be a vertex subset of G . If $G[S]$ has a cycle isolating set D such that each component of $G[S] - N[D]$ joins to $V(G) \setminus S$ by at most one edge, then $\iota_c(G) \leq |D| + \iota_c(G - S)$.

Corollary 2.2. Let G be a graph and S be a vertex subset of G . If D is a dominating set of $G[S]$, then $\iota_c(G) \leq |D| + \iota_c(G - S)$.

Proof. Since D is a dominating set of $G[S]$, we deduce that $S \setminus N[D] = \emptyset$ and thus $E(S \setminus N[D], V(G) \setminus S) = \emptyset$. Then by Lemma 2.1, we see that the assertion of the corollary holds. ■

The next lemma was proved by Borg in [2].

Lemma 2.3. (Borg [2]) If $G_1, G_2, \dots, G_s$ are the distinct components of a graph G , then $\iota_c(G) = \sum_{i=1}^s \iota_c(G_i)$.

The following observation is easy to verify, hence we omit the proof here.

Observation 2.4. The graphs in $\mathcal{G}_1$ satisfy the following properties:

- (i) for each $i \in [10]$, A_i is 2-connected;
- (ii) for each $i \in [10]$ and $j \in [8]$, $\{a_j, a_{9-j}\}$ is a cycle isolating set of A_i ;
- (iii) for each $i \in [10]$ and $j \in [8]$, there exists some $k \in [8]$ such that a_k has degree 4 in A_i and $a_j \notin N[a_k]$ unless one of the following holds:
 - (a) $i = 2$ and $j \in \{1, 5\}$;
 - (b) $i = 6$ and $j \in \{5, 6, 7, 8\}$;
 - (c) $i = 10$ and $j \in [8]$.

The final two lemmas show that every graph in $\mathcal{G}_1 \cup \mathcal{G}_2$ attains the bound stated in Theorem 1.3.

Lemma 2.5. If $G \in \mathcal{G}_1$, then $\iota_c(G) = 2 = \frac{|V(G)|}{4}$.

Proof. Since $G \in \mathcal{G}_1$, we have $G \cong A_i$ for some $i \in [10]$ and thus $|V(G)| = 8$. It is straightforward to verify that for any $j \in [8]$, $G - N[a_j]$ contains a C_3 or a C_4 . This implies that $\iota_c(G) \geq 2$. On the other hand, it follows from Theorem 1.3 that $\iota_c(G) \leq \frac{|V(G)|}{4} = 2$. Therefore, $\iota_c(G) = 2 = \frac{|V(G)|}{4}$. ■

Lemma 2.6. If $G \in \mathcal{G}_2$, then $\iota_c(G) = \frac{|V(G)|}{4}$.

Proof. Let $G_1, G_2, \dots, G_t$ be the C_3 -constituents of G , and for each $i \in [t]$, let u_i be the C_3 -connection of G_i and $H_i := u_i^1 u_i^2 u_i^3 u_i^1$ be the copy of C_3 in G_i . Then $|V(G)| = 4t$. It is easy to see that for any cycle isolating set D of G , we must have $D \cap V(G_i) \neq \emptyset$ for each $i \in [t]$; otherwise, some H_i would be a C_3 in $G - N[D]$, a contradiction. This means that $\iota_c(G) \geq t$. On the other hand, by Theorem 1.3, we conclude that $\iota_c(G) \leq \frac{|V(G)|}{4} = t$. Hence, $\iota_c(G) = t = \frac{|V(G)|}{4}$. ■

3. Proof of Theorem 1.4

In this section, we prove Theorem 1.4. Our proof is inductive and follows the lines of that in [2,6], but the arguments are much more complicated.

First, we prove two lemmas which verify the necessity of Theorem 1.4 when $n = 4$ or $n = 8$.

Lemma 3.1. Let G be a connected graph with 4 vertices. If $\iota_c(G) = 1$, then $G \in \{C_4\} \cup \mathcal{G}_2$.

Proof. Since $\iota_c(G) = 1$, G contains at least one cycle. This implies that $|E(G)| \geq 4$ (because G is connected). Then, one can easily check that $G \in \{C_4, C_3^+\}$ if $|E(G)| = 4$, $G \cong K_4^-$ if $|E(G)| = 5$ and $G \cong K_4$ if $|E(G)| = 6$. Since $\{C_3^+, K_4^-, K_4\} \subseteq \mathcal{G}_2$, we see that $G \in \{C_4, C_3^+, K_4^-, K_4\} \subseteq \{C_4\} \cup \mathcal{G}_2$. ■

Lemma 3.2. Let G be a connected graph with 8 vertices. If $\iota_c(G) = 2$, then $G \in \mathcal{G}_1 \cup \mathcal{G}_2$.

Proof. For the sake of convenience, we may assume that $V(G) = \{v_i : 1 \leq i \leq 8\}$. Since G is connected and $|V(G)| = 8$, we know that $\Delta(G) \geq 2$. If $\Delta(G) = 2$, then $G \in \{P_8, C_8\}$ and we derive a contradiction to the assumption that $\iota_c(G) = 2$ (because $\iota_c(P_8) = 0$ and $\iota_c(C_8) = 1$). On the other hand, if there exists some $i \in [8]$ such that $d(v_i) \geq 5$, then it follows from $|V(G)| = 8$ that $|V(G - N[v_i])| \leq 2$ and thus $\{v_i\}$ is a cycle isolating set of G , again contradicting the assumption that $\iota_c(G) = 2$. Therefore, $3 \leq \Delta(G) \leq 4$.

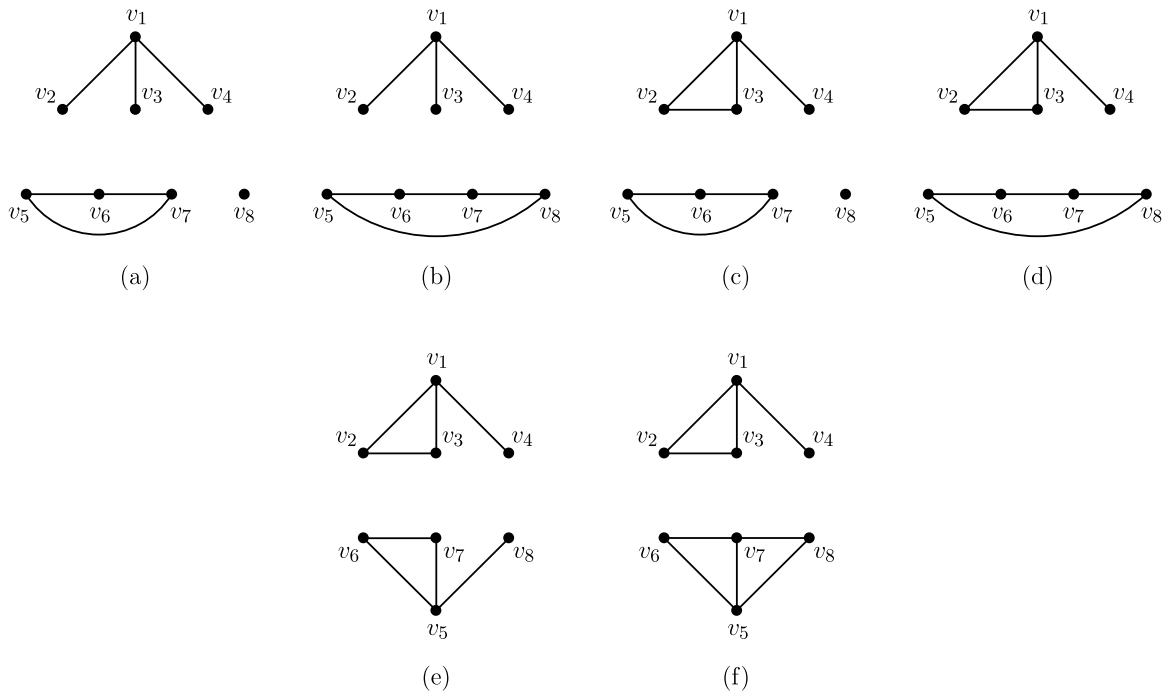


Fig. 3. The possible configurations in Subcases 1.1 and 1.2.

Case 1. $\Delta(G) = 3$.

Without loss of generality, we may assume that $d(v_1) = 3$ with $N(v_1) = \{v_2, v_3, v_4\}$. Define $G_1 := G[N[v_1]]$ and $G_2 := G - N[v_1]$. Since $\Delta(G) = 3$ and G is connected, $G_1 \in \{S_4, C_3^+, K_4^-\}$ and $G_2 \not\cong K_4$. If G_2 contains no cycle, then $\{v_1\}$ is a cycle isolating set of G , a contradiction. Hence, G_2 contains at least one cycle. This shows that $G_2 \in \{C_3 \cup K_1, C_4, C_3^+, K_4^-\}$. We consider three subcases according to the structure of G_1 .

Subcase 1.1. $G_1 \cong S_4$.

If $G_2 \in \{C_3^+, K_4^-\}$, then there must exist some $i \in \{5, 6, 7, 8\}$ such that v_i has degree 3 in G_2 and it is easy to observe that $\{v_i\}$ is a cycle isolating set of G , a contradiction. Thus, $G_2 \in \{C_3 \cup K_1, C_4\}$.

- Suppose $G_2 \cong C_3 \cup K_1$. Without loss of generality, we may assume that $v_5v_6, v_6v_7, v_7v_5 \in E(G)$ (see Fig. 3(a)). Since G is connected, we may further assume by symmetry that $v_2v_5 \in E(G)$. If $|E(\{v_8\}, \{v_3, v_4\})| \leq 1$, then $\{v_5\}$ is a cycle isolating set of G , a contradiction. Therefore, $|E(\{v_8\}, \{v_3, v_4\})| = 2$ (i.e. $v_8v_3, v_8v_4 \in E(G)$). This implies that each vertex in $\{v_3, v_4\}$ has at most one neighbor in $\{v_6, v_7\}$ (since $\Delta(G) = 3$). If $v_8v_2 \in E(G)$ or $E(\{v_3, v_4\}, \{v_6, v_7\}) = \emptyset$, then $\{v_2\}$ is a cycle isolating set of G , a contradiction. Hence, we have $v_8v_2 \notin E(G)$ and $E(\{v_3, v_4\}, \{v_6, v_7\}) \neq \emptyset$. By symmetry, we may assume that $v_3v_6 \in E(G)$. Now, it is easy to verify that $\{v_6\}$ is a cycle isolating set of G , giving a contradiction.
- Suppose $G_2 \cong C_4$. Without loss of generality, we may assume that $v_5v_6, v_6v_7, v_7v_8, v_8v_5 \in E(G)$ (see Fig. 3(b)). Then each vertex in $\{v_5, v_6, v_7, v_8\}$ has at most one neighbor in $\{v_2, v_3, v_4\}$ (because $\Delta(G) = 3$). Since G is connected, we may assume by symmetry that $v_2v_5 \in E(G)$. But then, $\{v_5\}$ is a cycle isolating set of G , a contradiction.

Subcase 1.2. $G_1 \cong C_3^+$.

Without loss of generality, suppose $v_2v_3 \in E(G)$ and $v_2v_4, v_3v_4 \notin E(G)$. Since $\Delta(G) = 3$, each vertex in $\{v_2, v_3\}$ has at most one neighbor in $\{v_5, v_6, v_7, v_8\}$. We consider all four possibilities of G_2 .

- Suppose $G_2 \cong C_3 \cup K_1$. Without loss of generality, we may assume that $v_5v_6, v_6v_7, v_7v_5 \in E(G)$ (see Fig. 3(c)). First, suppose $E(\{v_2, v_3\}, \{v_5, v_6, v_7\}) \neq \emptyset$. By symmetry, we may assume that $v_2v_5 \in E(G)$. If $|E(\{v_8\}, \{v_3, v_4\})| \leq 1$, then $\{v_5\}$ is a cycle isolating set of G , a contradiction. Thus, $|E(\{v_8\}, \{v_3, v_4\})| = 2$ (i.e. $v_8v_3, v_8v_4 \in E(G)$). This means that v_4 has at most one neighbor in $\{v_6, v_7\}$ (since $\Delta(G) = 3$). But now, $\{v_2\}$ is a cycle isolating set of G , again a contradiction. Next, suppose $E(\{v_2, v_3\}, \{v_5, v_6, v_7\}) = \emptyset$. Since G is connected, $E(\{v_4\}, \{v_5, v_6, v_7\}) \neq \emptyset$. If $v_8v_4 \in E(G)$ or $|E(\{v_8\}, \{v_2, v_3\})| \leq 1$, then $\{v_4\}$ is a cycle isolating set of G , a contradiction. Therefore, we can derive that $v_8v_2, v_8v_3 \in E(G)$ and $v_8v_4 \notin E(G)$. Define $X := \{v_1, v_2, v_3, v_8\}$ and $Y := \{v_4, v_5, v_6, v_7\}$. Then, it is easy to check that $G \in \mathcal{G}_2$ whose C_3 -constituents are $G[X]$ and $G[Y]$ (where $G[X] \cong K_4^-$ and $G[Y] \in \{C_3^+, K_4^-\}$).

- Suppose $G_2 \cong C_4$. Without loss of generality, we may assume that $v_5v_6, v_6v_7, v_7v_8, v_8v_5 \in E(G)$ (see Fig. 3(d)). Since $\Delta(G) = 3$, each vertex in $\{v_5, v_6, v_7, v_8\}$ has at most one neighbor in $\{v_2, v_3, v_4\}$. If $E(\{v_2, v_3\}, \{v_5, v_6, v_7, v_8\}) \neq \emptyset$, say $v_2v_5 \in E(G)$, then $\{v_5\}$ is a cycle isolating set of G , giving a contradiction. Hence, $E(\{v_2, v_3\}, \{v_5, v_6, v_7, v_8\}) = \emptyset$. This shows that v_4 has at least one neighbor in $\{v_5, v_6, v_7, v_8\}$ (since G is connected). But then, $\{v_4\}$ is a cycle isolating set of G , a contradiction.
- Suppose $G_2 \cong C_3^+$. Without loss of generality, we may assume that $v_5v_6, v_5v_7, v_5v_8, v_6v_7 \in E(G)$ (see Fig. 3(e)). First, suppose $E(\{v_2, v_3\}, \{v_6, v_7\}) \neq \emptyset$. By symmetry, we may assume that $v_2v_6 \in E(G)$. If $|E(\{v_8\}, \{v_3, v_4\})| \leq 1$, then $\{v_6\}$ is a cycle isolating set of G , a contradiction. Hence, $|E(\{v_8\}, \{v_3, v_4\})| = 2$ (i.e. $v_8v_3, v_8v_4 \in E(G)$). But then, $\{v_8\}$ is a cycle isolating set of G , giving a contradiction. Next, suppose $E(\{v_2, v_3\}, \{v_6, v_7\}) = \emptyset$. If $E(\{v_4\}, \{v_6, v_7\}) = E(\{v_8\}, \{v_2, v_3\}) = \emptyset$, then $v_4v_8 \in E(G)$ (since G is connected) and one can easily see that $G \in \mathcal{G}_2$ whose C_3 -constituents are G_1 and G_2 (where $G_1 \cong C_3^+$ and $G_2 \cong C_3^+$). Assume now by symmetry that $E(\{v_4\}, \{v_6, v_7\}) \neq \emptyset$, say $v_4v_6 \in E(G)$. If $|E(\{v_8\}, \{v_2, v_3\})| \leq 1$, then $\{v_4\}$ is a cycle isolating set of G , a contradiction. Therefore, we may further assume that $|E(\{v_8\}, \{v_2, v_3\})| = 2$ (i.e. $v_8v_2, v_8v_3 \in E(G)$). This implies that $v_4v_7 \in E(G)$; otherwise, $\{v_8\}$ is a cycle isolating set of G , a contradiction. Then, it is straightforward to verify that $G \cong A_{10} \in \mathcal{G}_1$.
- Suppose $G_2 \cong K_4^-$. Without loss of generality, we may assume that $v_5v_6, v_5v_7, v_5v_8, v_6v_7, v_7v_8 \in E(G)$ (see Fig. 3(f)). Since $\Delta(G) = 3$, each vertex in $\{v_5, v_7\}$ has no neighbor in $\{v_2, v_3, v_4\}$ and each vertex in $\{v_6, v_8\}$ has at most one neighbor in $\{v_2, v_3, v_4\}$. If $E(\{v_2, v_3\}, \{v_6, v_8\}) \neq \emptyset$, say $v_2v_6 \in E(G)$, then $\{v_6\}$ is a cycle isolating set of G , a contradiction. Thus, we have $E(\{v_2, v_3\}, \{v_6, v_8\}) = \emptyset$, and hence $E(\{v_4\}, \{v_6, v_8\}) \neq \emptyset$ (since G is connected). If $|E(\{v_4\}, \{v_6, v_8\})| = 1$, then $G \in \mathcal{G}_2$ whose C_3 -constituents are G_1 and G_2 (where $G_1 \cong C_3^+$ and $G_2 \cong K_4^-$). If $|E(\{v_4\}, \{v_6, v_8\})| = 2$ (i.e. $v_4v_6, v_4v_8 \in E(G)$), then $\{v_4\}$ is a cycle isolating set of G , a contradiction.

Subcase 1.3. $G_1 \cong K_4^-$.

Without loss of generality, suppose $v_2v_3, v_3v_4 \in E(G)$ and $v_2v_4 \notin E(G)$. Since $\Delta(G) = 3$, v_3 has no neighbor in $\{v_5, v_6, v_7, v_8\}$. This implies that $E(\{v_2, v_4\}, \{v_5, v_6, v_7, v_8\}) \neq \emptyset$ (because G is connected). By symmetry, we may assume that $E(\{v_2\}, \{v_5, v_6, v_7, v_8\}) \neq \emptyset$. Then $d(v_2) = 3$ and $G[N[v_2]] \cong C_3^+$, and we are back to Subcase 1.2 by relabeling v_2 as v_1 (and relabeling other vertices of G appropriately).

Case 2. $\Delta(G) = 4$.

Without loss of generality, we may assume that $d(v_1) = 4$ with $N(v_1) = \{v_2, v_3, v_4, v_5\}$. Define $G_1 := G[N[v_1]]$ and $G_2 := G - N[v_1]$. Then $G_2 \cong C_3$ (i.e. $v_6v_7, v_7v_8, v_8v_6 \in E(G)$); otherwise, $\{v_1\}$ is a cycle isolating set of G , a contradiction.

We consider the structure of G_1 . It is obvious that $G_1 \not\cong S_5$; otherwise, $\{v_6\}$ is a cycle isolating set of G , a contradiction. Moreover, since $\Delta(G) = 4$ and G is connected, $G_1 \not\cong K_5$ and thus $\delta(G_1) \leq 3$.

We further claim that $\delta(G_1) \leq 2$. Suppose to the contrary that $\delta(G_1) = 3$. Then each vertex in $\{v_2, v_3, v_4, v_5\}$ has at most one neighbor in $\{v_6, v_7, v_8\}$ (because $\Delta(G) = 4$). Since G is connected, there must exist some $i \in \{2, 3, 4, 5\}$ such that v_i has degree 3 in G_1 and $E(\{v_i\}, \{v_6, v_7, v_8\}) \neq \emptyset$. But then, it is easy to check that $\{v_i\}$ is a cycle isolating set of G , a contradiction.

Define $G'_1 := G[N(v_1)] = G_1 - v_1$. Then $\delta(G'_1) \leq 1$. This together with $G_1 \not\cong S_5$ shows that $G'_1 \in \{K_2 \cup 2K_1, P_3 \cup K_1, 2K_2, P_4, S_4, C_3 \cup K_1, C_3^+\}$. We now consider seven subcases according to the structure of G'_1 .

Subcase 2.1. $G'_1 \cong K_2 \cup 2K_1$.

Without loss of generality, we may assume that $v_2v_3 \in E(G)$ (see Fig. 4(a)). If $E(\{v_2, v_3\}, \{v_6, v_7, v_8\}) \neq \emptyset$, say $v_2v_6 \in E(G)$, then $\{v_6\}$ is a cycle isolating set of G , a contradiction. Therefore, we have $E(\{v_2, v_3\}, \{v_6, v_7, v_8\}) = \emptyset$. Since G is connected, $E(\{v_4, v_5\}, \{v_6, v_7, v_8\}) \neq \emptyset$. By symmetry, we may assume that $v_4v_6 \in E(G)$. But then, either $\{v_4\}$ (if $|E(\{v_5\}, \{v_7, v_8\})| \leq 1$) or $\{v_5\}$ (if $|E(\{v_5\}, \{v_7, v_8\})| = 2$) is a cycle isolating set of G , giving a contradiction.

Subcase 2.2. $G'_1 \cong P_3 \cup K_1$.

Without loss of generality, we may assume that $v_2v_3, v_3v_4 \in E(G)$ (see Fig. 4(b)). If there exists some $i \in \{6, 7, 8\}$ such that $v_iv_3 \in E(G)$ or $|E(\{v_i\}, \{v_2, v_4\})| = 2$ (i.e. $v_iv_2, v_iv_4 \in E(G)$), then $\{v_i\}$ is a cycle isolating set of G , a contradiction. Hence, we conclude that for each $i \in \{6, 7, 8\}$, $v_iv_3 \notin E(G)$ and $|E(\{v_i\}, \{v_2, v_4\})| \leq 1$. This implies that v_3 has no neighbor in $\{v_6, v_7, v_8\}$. Moreover, each vertex in $\{v_2, v_4, v_5\}$ has at most one neighbor in $\{v_6, v_7, v_8\}$; otherwise, suppose there exists some $j \in \{2, 4, 5\}$ such that v_j has at least two neighbors in $\{v_6, v_7, v_8\}$, then $\{v_j\}$ is a cycle isolating set of G , again a contradiction.

If $E(\{v_2, v_4\}, \{v_6, v_7, v_8\}) \neq \emptyset$, say $v_2v_6 \in E(G)$, then $\{v_2\}$ is a cycle isolating set of G , a contradiction. Thus, we have $E(\{v_2, v_4\}, \{v_6, v_7, v_8\}) = \emptyset$, and hence $E(\{v_5\}, \{v_6, v_7, v_8\}) \neq \emptyset$ (since G is connected). Now, it is easy to verify that $\{v_5\}$ is a cycle isolating set of G , giving a contradiction.

Subcase 2.3. $G'_1 \cong 2K_2$.

Without loss of generality, we may assume that $v_2v_3, v_4v_5 \in E(G)$ (see Fig. 4(c)). Since $\Delta(G) = 4$, each vertex in $\{v_2, v_3, v_4, v_5\}$ has at most two neighbors in $\{v_6, v_7, v_8\}$ and vice versa. If there exists some $i \in \{6, 7, 8\}$ such that $E(\{v_i\}, \{v_2, v_3\}) \neq \emptyset$ and $E(\{v_i\}, \{v_4, v_5\}) \neq \emptyset$, then $\{v_i\}$ is a cycle isolating set of G , a contradiction. Therefore,

(*) for each $i \in \{6, 7, 8\}$, either $E(\{v_i\}, \{v_2, v_3\}) = \emptyset$ or $E(\{v_i\}, \{v_4, v_5\}) = \emptyset$.

Suppose there exists some $i \in \{2, 3, 4, 5\}$ such that v_i has two neighbors in $\{v_6, v_7, v_8\}$. By symmetry, we may assume that $v_2v_6, v_2v_7 \in E(G)$. If $|E(\{v_8\}, \{v_4, v_5\})| \leq 1$, then $\{v_2\}$ is a cycle isolating set of G , a contradiction. Thus, $|E(\{v_8\}, \{v_4, v_5\})| = 2$ (i.e. $v_8v_4, v_8v_5 \in E(G)$). Note that $E(\{v_6, v_7\}, \{v_4, v_5\}) = \emptyset$ (by (*)). If $E(\{v_3\}, \{v_6, v_7\}) = \emptyset$, then one

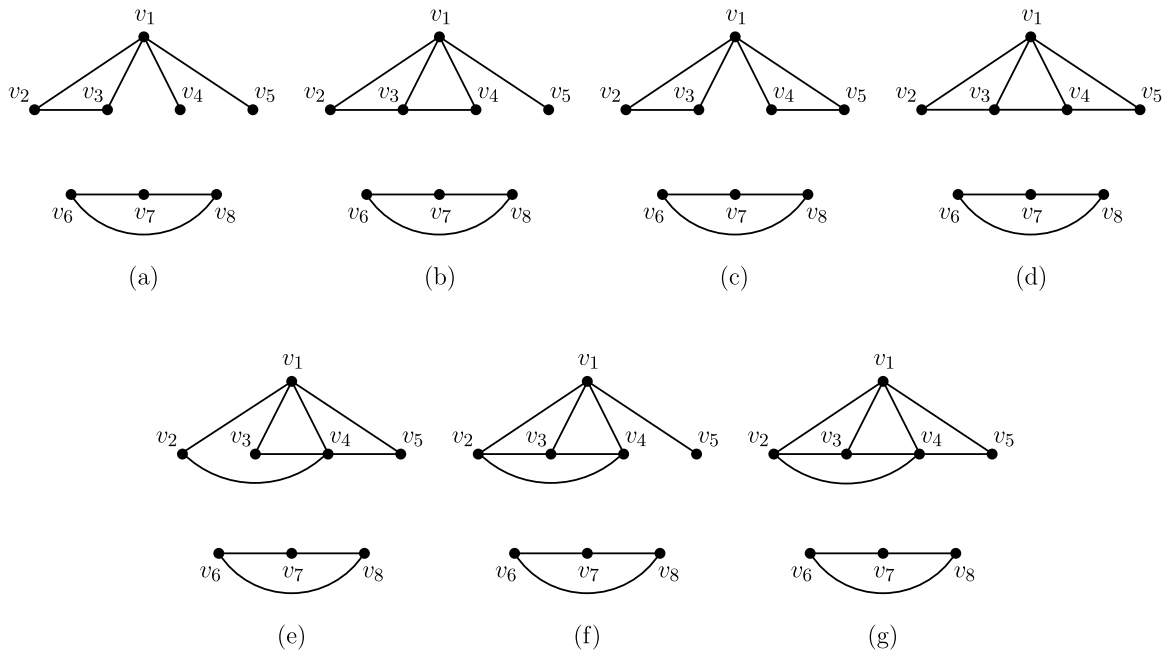


Fig. 4. The possible configurations in Case 2.

can easily check that $G \cong A_1 \in \mathcal{G}_1$. If $|E(\{v_3\}, \{v_6, v_7\})| = 1$, say $v_3v_6 \in E(G)$, then we can deduce that $G \cong A_3 \in \mathcal{G}_1$. If $|E(\{v_3\}, \{v_6, v_7\})| = 2$ (i.e. $v_3v_6, v_3v_7 \in E(G)$), then $G \cong A_4 \in \mathcal{G}_1$.

Now, assume that each vertex in $\{v_2, v_3, v_4, v_5\}$ has at most one neighbor in $\{v_6, v_7, v_8\}$. There are two possibilities to consider.

- Suppose there exists some $i \in \{6, 7, 8\}$ such that v_i has two neighbors in $\{v_2, v_3, v_4, v_5\}$. By (*) and by symmetry, we may assume that $v_6v_2, v_6v_3 \in E(G)$. Notice that $|E(\{v_4, v_5\}, \{v_7, v_8\})| \leq 2$. If $|E(\{v_4, v_5\}, \{v_7, v_8\})| \leq 1$, then $\{v_2\}$ is a cycle isolating set of G , a contradiction. Therefore, $|E(\{v_4, v_5\}, \{v_7, v_8\})| = 2$. First, suppose that the two edges in $E(\{v_4, v_5\}, \{v_7, v_8\})$ have a common endvertex. Without loss of generality, we may assume that $E(\{v_4, v_5\}, \{v_7, v_8\}) = \{v_4v_7, v_5v_7\}$. Then $E(\{v_7\}, \{v_2, v_3, v_4, v_5\}) = \emptyset$ (since each vertex in $\{v_2, v_3, v_4, v_5\}$ has at most one neighbor in $\{v_6, v_7, v_8\}$) and it is straightforward to verify that $G \cong A_1 \in \mathcal{G}_1$. Next, suppose that the two edges in $E(\{v_4, v_5\}, \{v_7, v_8\})$ have no common endvertex. Without loss of generality, we may assume that $E(\{v_4, v_5\}, \{v_7, v_8\}) = \{v_4v_7, v_5v_8\}$. Then, we can derive that $G \cong A_2 \in \mathcal{G}_1$.
- Suppose that each vertex in $\{v_6, v_7, v_8\}$ also has at most one neighbor in $\{v_2, v_3, v_4, v_5\}$. Since G is connected, we may assume by symmetry that $v_2v_6 \in E(G)$. Note that $|E(\{v_4, v_5\}, \{v_7, v_8\})| \leq 2$. If $|E(\{v_4, v_5\}, \{v_7, v_8\})| \leq 1$, then $\{v_2\}$ is a cycle isolating set of G , a contradiction. If $|E(\{v_4, v_5\}, \{v_7, v_8\})| = 2$, say $v_4v_7, v_5v_8 \in E(G)$, then $\{v_4\}$ is a cycle isolating set of G , again a contradiction.

Subcase 2.4. $G'_1 \cong P_4$.

Without loss of generality, we may assume that $v_2v_3, v_3v_4, v_4v_5 \in E(G)$ (see Fig. 4(d)).

First, suppose $E(\{v_3, v_4\}, \{v_6, v_7, v_8\}) \neq \emptyset$. By symmetry, we may assume that $v_3v_6 \in E(G)$. If $|E(\{v_5\}, \{v_7, v_8\})| \leq 1$, then $\{v_3\}$ is a cycle isolating set of G , a contradiction. Hence, $|E(\{v_5\}, \{v_7, v_8\})| = 2$ (i.e. $v_5v_7, v_5v_8 \in E(G)$). This shows that $v_2v_6 \in E(G)$; otherwise, $\{v_5\}$ is a cycle isolating set of G , again a contradiction. If $E(\{v_2\}, \{v_7, v_8\}) = \emptyset$, then $d(v_6) = 4$ and $G[N(v_6)] \cong 2K_2$, and we are back to Subcase 2.3 by relabeling v_6 as v_1 (and relabeling other vertices of G appropriately). Similarly, if $E(\{v_4\}, \{v_7, v_8\}) = \emptyset$, then $d(v_5) = 4$ and $G[N(v_5)] \cong 2K_2$, and we are back to Subcase 2.3 by relabeling v_5 as v_1 (and relabeling other vertices of G appropriately). Therefore, we may assume that for each $i \in \{2, 4\}$, $E(\{v_i\}, \{v_7, v_8\}) \neq \emptyset$. Since $\Delta(G) = 4$, we may further assume by symmetry that $v_2v_7, v_4v_8 \in E(G)$. Now, it is easy to verify that $G \cong A_5 \in \mathcal{G}_1$.

Next, suppose $E(\{v_3, v_4\}, \{v_6, v_7, v_8\}) = \emptyset$. Since G is connected, $E(\{v_2, v_5\}, \{v_6, v_7, v_8\}) \neq \emptyset$. By symmetry, we may assume that $v_2v_6 \in E(G)$. But then, either $\{v_2\}$ (if $|E(\{v_5\}, \{v_7, v_8\})| \leq 1$) or $\{v_5\}$ (if $|E(\{v_5\}, \{v_7, v_8\})| = 2$) is a cycle isolating set of G , a contradiction.

Subcase 2.5. $G'_1 \cong S_4$.

Without loss of generality, we may assume that $v_4v_2, v_4v_3, v_4v_5 \in E(G)$ (see Fig. 4(e)). Since $\Delta(G) = 4$, v_4 has no neighbor in $\{v_6, v_7, v_8\}$ and each vertex in $\{v_2, v_3, v_5\}$ has at most two neighbors in $\{v_6, v_7, v_8\}$. If there exists some $i \in \{2, 3, 5\}$ such that v_i has two neighbors in $\{v_6, v_7, v_8\}$, then $\{v_i\}$ is a cycle isolating set of G , a contradiction. Thus, we further see that each vertex in $\{v_2, v_3, v_5\}$ has at most one neighbor in $\{v_6, v_7, v_8\}$. Since G is connected, we may assume by symmetry that $v_2v_6 \in E(G)$. But then, it is easy to check that $\{v_2\}$ is a cycle isolating set of G , giving a contradiction.

Subcase 2.6. $G'_1 \cong C_3 \cup K_1$.

Without loss of generality, we may assume that $v_2v_3, v_3v_4, v_4v_2 \in E(G)$ (see Fig. 4(f)).

First, suppose $E(\{v_2, v_3, v_4\}, \{v_6, v_7, v_8\}) \neq \emptyset$. By symmetry, we may assume that $v_2v_6 \in E(G)$. Then $|E(\{v_5\}, \{v_7, v_8\})| = 2$ (i.e. $v_5v_7, v_5v_8 \in E(G)$); otherwise, $\{v_2\}$ is a cycle isolating set of G , a contradiction. If $E(\{v_2, v_3, v_4, v_5\}, \{v_6, v_7, v_8\}) = \{v_2v_6, v_5v_7, v_5v_8\}$, then we see that $G \cong A_6 \in \mathcal{G}_1$. Assume now that $|E(\{v_2, v_3, v_4, v_5\}, \{v_6, v_7, v_8\})| \geq 4$. We distinguish two possibilities according to whether $v_5v_6 \in E(G)$ or not.

- Suppose $v_5v_6 \in E(G)$. Note that $|E(\{v_3, v_4\}, \{v_7, v_8\})| \leq 2$. If $E(\{v_3, v_4\}, \{v_7, v_8\}) = \emptyset$, then one can easily check that $G \cong A_7 \in \mathcal{G}_1$. If $|E(\{v_3, v_4\}, \{v_7, v_8\})| = 1$, say $v_3v_7 \in E(G)$, then $G \cong A_8 \in \mathcal{G}_1$. If $|E(\{v_3, v_4\}, \{v_7, v_8\})| = 2$, say $v_3v_7, v_4v_8 \in E(G)$, then we can deduce that $G \cong A_9 \in \mathcal{G}_1$.
- Suppose $v_5v_6 \notin E(G)$. If $E(\{v_3, v_4\}, \{v_6\}) \neq \emptyset$, say $v_3v_6 \in E(G)$, then $\{v_6\}$ is a cycle isolating set of G , a contradiction. Hence, $E(\{v_3, v_4\}, \{v_6\}) = \emptyset$. Since $\Delta(G) = 4$ and $|E(\{v_2, v_3, v_4, v_5\}, \{v_6, v_7, v_8\})| \geq 4$, we conclude that $E(\{v_3, v_4\}, \{v_7, v_8\}) \neq \emptyset$. By symmetry, we may assume that $v_3v_7 \in E(G)$. But now, $\{v_3\}$ is a cycle isolating set of G , again a contradiction.

Next, suppose $E(\{v_2, v_3, v_4\}, \{v_6, v_7, v_8\}) = \emptyset$. Since G is connected, v_5 has at least one neighbor in $\{v_6, v_7, v_8\}$. Define $X := \{v_1, v_2, v_3, v_4\}$ and $Y := \{v_5, v_6, v_7, v_8\}$. Then, it is straightforward to verify that $G \in \mathcal{G}_2$ whose C_3 -constituents are $G[X]$ and $G[Y]$ (where $G[X] \cong K_4$ and $G[Y] \in \{C_3^+, K_4^-, K_4\}$).

Subcase 2.7. $G'_1 \cong C_3^+$.

Without loss of generality, we may assume that $v_2v_3, v_3v_4, v_4v_2, v_4v_5 \in E(G)$ (see Fig. 4(g)). Since $\Delta(G) = 4$, v_4 has no neighbor in $\{v_6, v_7, v_8\}$ and each vertex in $\{v_2, v_3\}$ has at most one neighbor in $\{v_6, v_7, v_8\}$.

First, suppose $E(\{v_2, v_3\}, \{v_6, v_7, v_8\}) \neq \emptyset$. By symmetry, we may assume that $v_2v_6 \in E(G)$. If $v_3v_6 \in E(G)$, then $d(v_6) = 4$ and $G[N(v_6)] \cong 2K_2$, and we are back to Subcase 2.3 by relabeling v_6 as v_1 (and relabeling other vertices of G appropriately). If $v_3v_6 \notin E(G)$, then $d(v_2) = 4$ and $G[N(v_2)] \cong C_3 \cup K_1$, and we are back to Subcase 2.6 by relabeling v_2 as v_1 (and relabeling other vertices of G appropriately).

Next, suppose $E(\{v_2, v_3\}, \{v_6, v_7, v_8\}) = \emptyset$. Since G is connected, v_5 has at least one neighbor in $\{v_6, v_7, v_8\}$. But then, it is easy to check that $\{v_5\}$ is a cycle isolating set of G , giving a contradiction.

This completes the proof of Lemma 3.2. ■

We are now ready to prove Theorem 1.4.

Proof of Theorem 1.4. If $G \cong C_4$, then $n = 4$ and $\iota_c(G) = 1 = \frac{n}{4}$. If $G \in \mathcal{G}_1 \cup \mathcal{G}_2$, then by Lemmas 2.5 and 2.6, we also have $\iota_c(G) = \frac{n}{4}$.

In the following, we shall prove the necessity by induction on n . Since $\iota_c(G) = \frac{n}{4}$, n is a multiple of 4. If $n = 4$, then it follows from Lemma 3.1 that $G \in \{C_4\} \cup \mathcal{G}_2$. If $n = 8$, then by Lemma 3.2, we also derive that $G \in \mathcal{G}_1 \cup \mathcal{G}_2$. So we may assume that $n \geq 12$.

If $\Delta(G) = 2$, then $G \in \{P_n, C_n\}$. Since $\iota_c(P_n) = 0$ and $\iota_c(C_n) = 1$, we have $\iota_c(G) \leq 1 < \frac{n}{4}$ when $n \geq 12$, which contradicts the assumption that $\iota_c(G) = \frac{n}{4}$.

Therefore, $\Delta(G) \geq 3$. Let v be a vertex in G such that $d(v) = \Delta(G)$. Then $V(G) \neq N[v]$; otherwise, $\{v\}$ is a cycle isolating set of G and thus $\iota_c(G) \leq 1 < \frac{n}{4}$ (since $n \geq 12$), a contradiction. Define $G' := G - N[v]$. Let $\mathcal{H}$ be the set of components of G' . For any $x \in N(v)$ and any $H \in \mathcal{H}$, we say that x is *linked to* H or H is *linked to* x if x has at least one neighbor in H . Let $\mathcal{H}_b := \{H \in \mathcal{H} : H \cong C_3\}$ and $\mathcal{H}_g := \mathcal{H} \setminus \mathcal{H}_b$. Then by Theorem 1.3 and the induction hypothesis, we see that for each $H \in \mathcal{H}_g$, $\iota_c(H) \leq \frac{|V(H)|}{4}$ with equality if and only if $H \in \{C_4\} \cup \mathcal{G}_1 \cup \mathcal{G}_2$. We consider two cases according to whether $\mathcal{H}_b = \emptyset$ or not.

Case 1. $\mathcal{H}_b = \emptyset$.

Note that $\{v\}$ is a dominating set of $G[N[v]]$. By Corollary 2.2, Lemma 2.3 and Theorem 1.3, we have

$$\iota_c(G) \leq |\{v\}| + \iota_c(G') = 1 + \sum_{H \in \mathcal{H}_g} \iota_c(H) \leq 1 + \sum_{H \in \mathcal{H}_g} \frac{|V(H)|}{4} = 1 + \frac{n-1-\Delta(G)}{4} = \frac{n}{4} - \frac{\Delta(G)-3}{4} \leq \frac{n}{4}. \quad (1)$$

Since $\iota_c(G) = \frac{n}{4}$, we know that all inequalities in (1) must be equalities, which implies that $\Delta(G) = 3$ and $\iota_c(H) = \frac{|V(H)|}{4}$ for each $H \in \mathcal{H}_g$. Then by the induction hypothesis, $H \in \{C_4\} \cup \mathcal{G}_1 \cup \mathcal{G}_2$ for each $H \in \mathcal{H}_g$. In the following two claims, we further show that $H \in \mathcal{G}_2$ for each $H \in \mathcal{H}_g$.

Claim 1. $H \notin \mathcal{G}_1$ for any $H \in \mathcal{H}_g$.

Proof. Let H be an arbitrary member of $\mathcal{H}_g$. Because $\Delta(G) = 3$ and H is linked to at least one vertex in $N(v)$, we have $\delta(H) \leq 2$. Since $\Delta(A_i) = 4$ for each $i \in [9]$ and A_{10} is cubic, $H \not\cong A_i$ for any $i \in [10]$, i.e. $H \notin \mathcal{G}_1$. ■

Claim 2. $H \not\cong C_4$ for any $H \in \mathcal{H}_g$.

Proof. Suppose to the contrary that there exists a member $H' \in \mathcal{H}_g$ such that $H' \cong C_4$. Define $X := N[v] \cup V(H')$. Then $|X| = 8$ and the components of $G - X$ are the members of $\mathcal{H}_g \setminus \{H'\}$. Since $n \geq 12$ and $|X| = 8$, we derive that $V(G) \setminus X \neq \emptyset$ (i.e. $\mathcal{H}_g \setminus \{H'\} \neq \emptyset$).

We first consider $G - X$. For each $H \in \mathcal{H}_g \setminus \{H'\}$, we define a cycle isolating set D_H of H as follows:

- If $H \cong C_4$, then let $D_H := \{x_H\}$, where x_H is an arbitrary vertex in H . It is clear that $H - N[D_H] \cong K_1$, and hence D_H is a cycle isolating set of H with $|D_H| = 1 = \frac{|V(H)|}{4}$. Moreover, since $\Delta(G) = 3$ and $H \cong C_4$, the remaining vertex of $H - N[D_H]$ joins to X by at most one edge.
- If $H \in \mathcal{G}_2$, then let $G_1, G_2, \dots, G_t$ be the C_3 -constituents of H , and for each $i \in [t]$, let u_i be the C_3 -connection of G_i and $H_i := u_i^1 u_i^2 u_i^3 u_i^1$ be the copy of C_3 in G_i . Define $D_H := \{u_i^1 : 1 \leq i \leq t\}$. Then, it is easy to observe that D_H is a dominating set (and thus a cycle isolating set) of H and $|D_H| = t = \frac{|V(H)|}{4}$.

Define $D := \bigcup_{H \in \mathcal{H}_g \setminus \{H'\}} D_H$. Then by Lemma 2.3, we see that D is a cycle isolating set of $G - X$ with

$$|D| = \sum_{H \in \mathcal{H}_g \setminus \{H'\}} \frac{|V(H)|}{4} = \frac{n-8}{4} = \frac{n}{4} - 2,$$

and each component of $G - X - N[D]$ joins to X by at most one edge.

We then consider $G[X]$. Since $E(N(v), V(H')) \neq \emptyset$, we know that $G[X]$ is connected. Moreover, $G[X] \notin \{C_3, C_4\}$ (because $|X| = 8$). Since G is connected and $V(G) \setminus X \neq \emptyset$, we deduce that $E(X, V(G) \setminus X) \neq \emptyset$ and hence $\delta(G[X]) \leq 2$. Because $\Delta(A_i) = 4$ for each $i \in [9]$ and A_{10} is cubic, $G[X] \not\cong A_i$ for any $i \in [10]$, i.e. $G[X] \notin \mathcal{G}_1$. On the other hand, since $G[X]$ contains an induced C_4 and no 8-vertex graph in $\mathcal{G}_2$ contains an induced C_4 , we further derive that $G[X] \notin \mathcal{G}_2$. Then by Theorem 1.3 and the induction hypothesis, we have $\iota_c(G[X]) < \frac{|X|}{4} = 2$.

Now, it follows from Lemma 2.1 that

$$\iota_c(G) \leq |D| + \iota_c(G[X]) < \left(\frac{n}{4} - 2\right) + 2 = \frac{n}{4},$$

contradicting the assumption that $\iota_c(G) = \frac{n}{4}$. ■

By Claims 1 and 2, $H \in \mathcal{G}_2$ for each $H \in \mathcal{H}_g$. Let $H_1, H_2, \dots, H_s$ be the members of $\mathcal{H}_g$. For each $i \in [s]$, let $G_{i,1}, G_{i,2}, \dots, G_{i,t_i}$ be the C_3 -constituents of H_i , and for each $j \in [t_i]$, let $u_{i,j}$ be the C_3 -connection of $G_{i,j}$ and $H_{i,j} := u_{i,j}^1 u_{i,j}^2 u_{i,j}^3 u_{i,j}^1$ be the copy of C_3 in $G_{i,j}$. For each $i \in [s]$ and $j \in [t_i]$, since G is connected and $\Delta(G) = 3$, we know that $\delta(G_{i,j}) \leq 2$ and thus $G_{i,j} \in \{C_3^+, K_4^-\}$.

Define $D_1 := \{u_{i,j}^1 : 1 \leq i \leq s \text{ and } 1 \leq j \leq t_i\}$ and $D_2 := \{u_{i,j} : 1 \leq i \leq s \text{ and } 1 \leq j \leq t_i\}$. Then, it is easy to verify that both D_1 and D_2 are cycle isolating sets of G' and

$$|D_1| = |D_2| = \frac{n-4}{4} = \frac{n}{4} - 1 = \iota_c(G) - 1.$$

Moreover, D_1 is a dominating set of G' . Let $N(v) = \{x_1, x_2, x_3\}$. If $G[N[v]] \cong S_4$, then one can easily check that D_1 is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. On the other hand, since G is connected, we also have $G[N[v]] \not\cong K_4$. Therefore, $G[N[v]] \in \{C_3^+, K_4^-\}$.

For each $i \in [s]$, since H_i is linked to at least one vertex in $N(v)$, there must exist at least one $j \in [t_i]$ such that $E(N(v), V(G_{i,j})) \neq \emptyset$. We now arbitrarily choose a C_3 -constituent $G_{i,j}$ with $E(N(v), V(G_{i,j})) \neq \emptyset$ (where $i \in [s]$ and $j \in [t_i]$). Define $Y := N[v] \cup V(G_{i,j})$. Then $|Y| = 8$ and $G[Y]$ is connected (and hence $G[Y] \not\cong C_3$). By Theorem 1.3, $\iota_c(G[Y]) \leq \frac{|Y|}{4} = 2$. On the other hand, it is easy to see that $D'_1 := D_1 \setminus \{u_{i,j}^1\}$ is a dominating set of $G - Y$ with $|D'_1| = |D_1| - 1 = \frac{n}{4} - 2$. Then by Corollary 2.2, we derive that

$$\iota_c(G) \leq |D'_1| + \iota_c(G[Y]) \leq \left(\frac{n}{4} - 2\right) + 2 = \frac{n}{4}. \quad (2)$$

Since $\iota_c(G) = \frac{n}{4}$, we see that all inequalities in (2) must be equalities, which implies that $\iota_c(G[Y]) = 2$. Note that $V(G) \setminus Y \neq \emptyset$ (because $n \geq 12$ and $|Y| = 8$). Since G is connected, we know that $E(Y, V(G) \setminus Y) \neq \emptyset$ and thus $\delta(G[Y]) \leq 2$.

We consider two subcases according to the structure of $G[N[v]]$.

Subcase 1.1. $G[N[v]] \cong C_3^+$.

Without loss of generality, we may assume that $x_1 x_2 \in E(G)$ and $x_1 x_3, x_2 x_3 \notin E(G)$. Since $\Delta(G) = 3$, $u_{i,j}^1$ has no neighbor in $\{x_1, x_2, x_3\}$. If $t_i = 1$, $H_i \cong K_4^-$ and $E(N(v), V(H_i)) = \{x_3 u_{i,1}^3\}$, then we exchange the labels of $u_{i,1}$ and $u_{i,1}^3$ (note that in this situation, the two vertices $u_{i,1}$ and $u_{i,1}^3$ are symmetric in H_i). After this modification, we shall show that $E(N(v), V(G_{i,j})) = \{x_3 u_{i,j}\}$.

First, suppose that x_3 has two neighbors in $\{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}$. If $x_3 u_{i,j}, x_3 u_{i,j}^k \in E(G)$ for some $k \in \{2, 3\}$, then $\{x_3\}$ is a cycle isolating set of $G[Y]$, which contradicts the fact that $\iota_c(G[Y]) = 2$. Hence, $x_3 u_{i,j}^2, x_3 u_{i,j}^3 \in E(G)$. If $|E(\{u_{i,j}\}, \{x_1, x_2\})| \leq 1$, then $\{x_3\}$ is still a cycle isolating set of $G[Y]$, again a contradiction. Thus, $|E(\{u_{i,j}\}, \{x_1, x_2\})| = 2$ (i.e. $x_1 u_{i,j}, x_2 u_{i,j} \in E(G)$). But then, we conclude that $G[Y]$ is cubic, contradicting the fact that $\delta(G[Y]) \leq 2$.

Next, suppose that x_3 has at most one neighbor in $\{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}$. If $u_{i,j}$ has two neighbors in $\{x_1, x_2, x_3\}$, then $\{u_{i,j}\}$ is a cycle isolating set of $G[Y]$, a contradiction. Therefore, $u_{i,j}$ has at most one neighbor in $\{x_1, x_2, x_3\}$. There are two possibilities to consider.

- Suppose $E(\{x_1, x_2\}, \{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}) \neq \emptyset$. If $E(\{x_1, x_2\}, \{u_{i,j}^2, u_{i,j}^3\}) \neq \emptyset$, say $x_1 u_{i,j}^k \in E(G)$ for some $k \in \{2, 3\}$, then $\{u_{i,j}^k\}$ is a cycle isolating set of $G[Y]$, a contradiction. Hence, we see that $E(\{x_1, x_2\}, \{u_{i,j}^2, u_{i,j}^3\}) = \emptyset$, and thus $E(\{x_1, x_2\}, \{u_{i,j}\}) \neq \emptyset$. But now, it is straightforward to verify that $\{u_{i,j}\}$ is a cycle isolating set of $G[Y]$, a contradiction.
- Suppose $E(\{x_1, x_2\}, \{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}) = \emptyset$. Since $G[Y]$ is connected, $E(\{x_3\}, \{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}) \neq \emptyset$. This (together with the assumption that x_3 has at most one neighbor in $\{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}$) means that $|E(\{x_3\}, \{u_{i,j}, u_{i,j}^2, u_{i,j}^3\})| = 1$. If $E(\{x_3\}, \{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}) = \{x_3 u_{i,j}^k\}$ for some $k \in \{2, 3\}$, then it is easy to check that $(D_2 \setminus \{u_{i,j}\}) \cup \{x_3\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, giving a contradiction. Thus, $E(\{x_3\}, \{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}) = \{x_3 u_{i,j}\}$, as desired.

By the choice of $G_{i,j}$, we can conclude that $E(N(v), V(G_{i,j})) = \{x_3 u_{i,j}\}$ for each $i \in [s]$ and $j \in [t_i]$ where $E(N(v), V(G_{i,j})) \neq \emptyset$. Recall that for each $i \in [s]$, there exists at least one $j \in [t_i]$ such that $E(N(v), V(G_{i,j})) \neq \emptyset$. Since $\Delta(G) = 3$, $s = |\mathcal{H}_g| \leq 2$. Define $D^* := \{x_3\} \cup D_2$. Then we derive that $G[D^*]$ is connected, and $G \in \mathcal{G}_2$ whose C_3 -constituents are $G[N[v]]$, $G_{1,1}$, $G_{1,2}$, $\dots$, G_{1,t_1} , $\dots$, $G_{s,1}$, $G_{s,2}$, $\dots$, G_{s,t_s} .

Subcase 1.2. $G[N[v]] \cong K_4^-$.

Without loss of generality, we may assume that $x_1 x_2, x_2 x_3 \in E(G)$ and $x_1 x_3 \notin E(G)$. Since $\Delta(G) = 3$, $u_{i,j}^1$ has no neighbor in $\{x_1, x_2, x_3\}$ and x_2 has no neighbor in $\{u_{i,j}, u_{i,j}^2, u_{i,j}^3\}$. Moreover, because $E(Y, V(G) \setminus Y) \neq \emptyset$, at least one vertex in $\{x_1, x_3, u_{i,j}\}$ has degree at most 2 in $G[Y]$. If $t_i = 1$, $H_i \cong K_4^-$ and $E(N(v), V(H_i)) = \{x_k u_{i,1}^3\}$ for some $k \in \{1, 3\}$, then we exchange the labels of $u_{i,1}$ and $u_{i,1}^3$ (note that in this situation, the two vertices $u_{i,1}$ and $u_{i,1}^3$ are symmetric in H_i). After this modification, we shall show that $E(N(v), V(G_{i,j})) = \{x_k u_{i,j}\}$ for some $k \in \{1, 3\}$.

First, suppose $E(\{x_1, x_3\}, \{u_{i,j}^2, u_{i,j}^3\}) \neq \emptyset$. By symmetry, we may assume that $E(\{x_3\}, \{u_{i,j}^2, u_{i,j}^3\}) \neq \emptyset$. Since at least one vertex in $\{x_1, x_3, u_{i,j}\}$ has degree at most 2 in $G[Y]$, we observe that $x_1 u_{i,j} \notin E(G)$ when $G_{i,j} \cong K_4^-$. Then, it is easy to verify that $(D_2 \setminus \{u_{i,j}\}) \cup \{x_3\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction.

Next, suppose $E(\{x_1, x_3\}, \{u_{i,j}^2, u_{i,j}^3\}) = \emptyset$. Then, it follows from $G[Y]$ is connected that $E(\{x_1, x_3\}, \{u_{i,j}\}) \neq \emptyset$. On the other hand, since at least one vertex in $\{x_1, x_3, u_{i,j}\}$ has degree at most 2 in $G[Y]$, we deduce that $|E(\{x_1, x_3\}, \{u_{i,j}\})| \leq 1$. Therefore, $|E(\{x_1, x_3\}, \{u_{i,j}\})| = 1$, as desired.

By the choice of $G_{i,j}$, for each $i \in [s]$ and $j \in [t_i]$ with $E(N(v), V(G_{i,j})) \neq \emptyset$, we always have $E(N(v), V(G_{i,j})) = \{x_k u_{i,j}\}$ for some $k \in \{1, 3\}$. If there exist two distinct C_3 -constituents G_{i_1,j_1} and G_{i_2,j_2} such that $E(N(v), V(G_{i_1,j_1})) = \{x_1 u_{i_1,j_1}\}$ and $E(N(v), V(G_{i_2,j_2})) = \{x_3 u_{i_2,j_2}\}$ (where $i_1, i_2 \in [s]$, $j_1 \in [t_{i_1}]$ and $j_2 \in [t_{i_2}]$), then one can easily check that D_2 is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. Since $\Delta(G) = 3$, there exists a unique C_3 -constituent $G_{i,j}$ with $E(N(v), V(G_{i,j})) \neq \emptyset$ (where $i \in [s]$ and $j \in [t_i]$). By symmetry, we may assume that $E(N(v), V(G_{i,j})) = \{x_3 u_{i,j}\}$. Recall that for each $i \in [s]$, there exists at least one $j \in [t_i]$ such that $E(N(v), V(G_{i,j})) \neq \emptyset$. Hence, $s = |\mathcal{H}_g| = 1$. Define $D^* := \{x_3\} \cup D_2$. Then we know that $G[D^*]$ is connected, and $G \in \mathcal{G}_2$ whose C_3 -constituents are $G[N[v]]$, $G_{1,1}$, $G_{1,2}$, $\dots$, G_{1,t_1} .

Case 2. $\mathcal{H}_b \neq \emptyset$.

For any given $x \in N(v)$, let $\mathcal{H}_b^x := \{H \in \mathcal{H}_b : H \text{ is linked to } x \text{ only}\}$ and $\mathcal{H}_g^x := \{H \in \mathcal{H}_g : H \text{ is linked to } x \text{ only}\}$. We have two subcases to consider.

Subcase 2.1. There exists a vertex $x \in N(v)$ such that $\mathcal{H}_b^x \neq \emptyset$.

For the sake of brevity, we write $b := |\mathcal{H}_b^x|$. Since $\mathcal{H}_b^x \neq \emptyset$, we have $b \geq 1$. Let $X := \{x\} \cup \left(\bigcup_{H \in \mathcal{H}_b^x} V(H)\right)$ and let G_v be the component of $G - X$ containing v . Then the components of $G - X$ are G_v and the members of $\mathcal{H}_g^x$. We consider two subcases according to whether $G_v \cong C_3$ or not.

Subcase 2.1.1. $G_v \not\cong C_3$.

Note that $\{x\}$ is a cycle isolating set of $G[X]$ and $E(X \setminus N[x], V(G) \setminus X) = \emptyset$. By [Lemmas 2.1, 2.3](#) and [Theorem 1.3](#), we deduce that

$$\iota_c(G) \leq |\{x\}| + \iota_c(G - X) = 1 + \sum_{H \in \mathcal{H}_g^x \cup \{G_v\}} \iota_c(H) \leq 1 + \sum_{H \in \mathcal{H}_g^x \cup \{G_v\}} \frac{|V(H)|}{4} = 1 + \frac{n - 1 - 3b}{4} = \frac{n}{4} - \frac{3b - 3}{4} \leq \frac{n}{4}. \quad (3)$$

Since $\iota_c(G) = \frac{n}{4}$, we conclude that all inequalities in (3) must be equalities, which shows that $b = 1$ and $\iota_c(H) = \frac{|V(H)|}{4}$ for each $H \in \mathcal{H}_g^x \cup \{G_v\}$. Then by the induction hypothesis, $H \in \{C_4\} \cup \mathcal{G}_1 \cup \mathcal{G}_2$ for each $H \in \mathcal{H}_g^x \cup \{G_v\}$. Moreover, it follows from $b = 1$ that $G[X] \in \{C_3^+, K_4^-, K_4\} \subseteq \mathcal{G}_2$ with x as its C_3 -connection.

Claim 3. $H \in \mathcal{G}_2$ for every $H \in \mathcal{H}_g^x \cup \{G_v\}$.

Proof. Suppose to the contrary that there exists a member $H' \in \mathcal{H}_g^x \cup \{G_v\}$ such that $H' \in \{C_4\} \cup \mathcal{G}_1$. Let w be a neighbor of x in H' . Define $Y := X \cup \{w\}$ and $H^* := H' - w$. Then, it is easy to observe that $\{x\}$ is a cycle isolating set of $G[Y]$ and $E(Y \setminus N[x], V(G) \setminus Y) = \emptyset$. Since C_4 is 2-connected and by [Observation 2.4\(i\)](#), we derive that H^* is connected. This implies that the components of $G - Y$ are H^* and the members of $(\mathcal{H}_g^x \cup \{G_v\}) \setminus \{H'\}$. Moreover, $H^* \not\cong C_3$ (since $H^* \cong P_3$ when $H' \cong C_4$ and $|V(H^*)| = 7$ when $H' \in \mathcal{G}_1$). By [Lemmas 2.1, 2.3](#) and [Theorem 1.3](#), we know that

$$\iota_c(G) \leq |\{x\}| + \iota_c(G - Y) = 1 + \iota_c(H^*) + \sum_{H \in (\mathcal{H}_g^x \cup \{G_v\}) \setminus \{H'\}} \iota_c(H)$$

$$\leq 1 + \frac{|V(H^*)|}{4} + \sum_{H \in (\mathcal{H}_g^x \cup \{G_v\}) \setminus \{H'\}} \frac{|V(H)|}{4} = 1 + \frac{n-5}{4} = \frac{n}{4} - \frac{1}{4} < \frac{n}{4},$$

a contradiction. ■

By Claim 3, let $H_1, H_2, \dots, H_s$ be the members of $\mathcal{H}_g^x \cup \{G_v\}$. For each $i \in [s]$, let $G_{i,1}, G_{i,2}, \dots, G_{i,t_i}$ be the C_3 -constituents of H_i , and for each $j \in [t_i]$, let $u_{i,j}$ be the C_3 -connection of $G_{i,j}$ and $H_{i,j} := u_{i,j}^1 u_{i,j}^2 u_{i,j}^3$ be the copy of C_3 in $G_{i,j}$. Recall that $G_{i,j} \in \{C_3^+, K_4^-, K_4\}$ for each $i \in [s]$ and $j \in [t_i]$.

Define $D := \{x\} \cup \{u_{i,j} : 1 \leq i \leq s \text{ and } 1 \leq j \leq t_i\}$. Then, it is straightforward to verify that D is a cycle isolating set of G and

$$|D| = 1 + \frac{n-4}{4} = \frac{n}{4} = \iota_c(G).$$

For each $i \in [s]$, we exchange the labels of vertices in H_i if one of the following holds:

- If $t_i = 1$, $H_i \cong K_4^-$ and $u_{i,1}^3$ is the unique neighbor of x in H_i , then we exchange the labels of $u_{i,1}$ and $u_{i,1}^3$ (note that in this situation, the two vertices $u_{i,1}$ and $u_{i,1}^3$ are symmetric in H_i).
- If $t_i = 1$, $H_i \cong K_4$ and there exists some $k \in [3]$ such that $u_{i,1}^k$ is the unique neighbor of x in H_i , then we exchange the labels of $u_{i,1}$ and $u_{i,1}^k$ (notice that in this situation, the four vertices $u_{i,1}, u_{i,1}^1, u_{i,1}^2$ and $u_{i,1}^3$ are symmetric in H_i).

After these modifications, let w_i be a neighbor of x in H_i for each $i \in [s]$. If, for some $i \in [s]$, w_i can be chosen such that $w_i = u_{i,j}^k$ for some $j \in [t_i]$ and $k \in [3]$, then it is easy to check that $D \setminus \{u_{i,j}\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. Thus, we can deduce that for each $i \in [s]$, $w_i = u_{i,j}$ for some $j \in [t_i]$. This means that $G[D]$ is connected, and $G \in \mathcal{G}_2$ whose C_3 -constituents are $G[X], G_{1,1}, G_{1,2}, \dots, G_{1,t_1}, \dots, G_{s,1}, G_{s,2}, \dots, G_{s,t_s}$.

Subcase 2.1.2. $G_v \cong C_3$.

Define $Y := X \cup V(G_v)$. Then the components of $G - Y$ are the members of $\mathcal{H}_g^x$. It is clear that $\{x\}$ is a cycle isolating set of $G[Y]$ and $E(Y \setminus N[x], V(G) \setminus Y) = \emptyset$. By Lemmas 2.1, 2.3 and Theorem 1.3, we have

$$\iota_c(G) \leq |\{x\}| + \iota_c(G - Y) = 1 + \sum_{H \in \mathcal{H}_g^x} \iota_c(H) \leq 1 + \sum_{H \in \mathcal{H}_g^x} \frac{|V(H)|}{4} = 1 + \frac{n-4-3b}{4} = \frac{n}{4} - \frac{3b}{4} < \frac{n}{4},$$

giving a contradiction.

Subcase 2.2. $\mathcal{H}_b^x = \emptyset$ for every $x \in N(v)$.

By the definition of $\mathcal{H}_b^x$, we conclude that each member of $\mathcal{H}_b$ is linked to at least two vertices in $N(v)$. We choose a vertex $x \in N(v)$ and a member $H^* \in \mathcal{H}_b$ such that H^* is linked to x . For the sake of convenience, we may assume that $H^* := y_1 y_2 y_3 y_1$ and $y_1 x \in E(G)$.

Let $X := \{x\} \cup V(H^*)$ and let G_v be the component of $G - X$ containing v . Then the components of $G - X$ are G_v and the members of $\mathcal{H}_g^x$. We consider two subcases according to whether $G_v \cong C_3$ or not.

Subcase 2.2.1. $G_v \not\cong C_3$.

Note that $\{y_1\}$ is a dominating set of $G[X]$. By Corollary 2.2, Lemma 2.3 and Theorem 1.3, we see that

$$\iota_c(G) \leq |\{y_1\}| + \iota_c(G - X) = 1 + \sum_{H \in \mathcal{H}_g^x \cup \{G_v\}} \iota_c(H) \leq 1 + \sum_{H \in \mathcal{H}_g^x \cup \{G_v\}} \frac{|V(H)|}{4} = 1 + \frac{n-4}{4} = \frac{n}{4}. \quad (4)$$

Since $\iota_c(G) = \frac{n}{4}$, we derive that all inequalities in (4) must be equalities, which implies that $\iota_c(H) = \frac{|V(H)|}{4}$ for each $H \in \mathcal{H}_g^x \cup \{G_v\}$. Then by the induction hypothesis, $H \in \{C_4\} \cup \mathcal{G}_1 \cup \mathcal{G}_2$ for each $H \in \mathcal{H}_g^x \cup \{G_v\}$.

Claim 4. $\mathcal{H}_g^x = \emptyset$.

Proof. Suppose to the contrary that $\mathcal{H}_g^x \neq \emptyset$. Let H' be a member of $\mathcal{H}_g^x$ and let w be a neighbor of x in H' . Then $H' \in \{C_4\} \cup \mathcal{G}_1 \cup \mathcal{G}_2$. We define a cycle isolating set $D_{H'}$ of H' as follows:

- If $H' \cong C_4$, then let $D_{H'} := \{w\}$. It is obvious that $D_{H'}$ is a cycle isolating set of H' and $|D_{H'}| = 1 = \frac{|V(H')|}{4}$.
- If $H' \in \mathcal{G}_1$, then $H' \cong A_i$ for some $i \in [10]$ (where the vertices of H' are labeled as shown in Fig. 2) and $w = a_j$ for some $j \in [8]$. Define $D_{H'} := \{w, a_{9-j}\}$. By Observation 2.4(ii), we deduce that $D_{H'}$ is a cycle isolating set of H' and $|D_{H'}| = 2 = \frac{|V(H')|}{4}$.
- If $H' \in \mathcal{G}_2$, then let $G_1, G_2, \dots, G_t$ be the C_3 -constituents of H' , and for each $i \in [t]$, let u_i be the C_3 -connection of G_i and $H_i := u_i^1 u_i^2 u_i^3$ be the copy of C_3 in G_i . Then $w \in V(G_i)$ for some $i \in [t]$. Define $D_{H'} := \{w\} \cup \{u_j : 1 \leq j \leq t \text{ and } j \neq i\}$. Then, one can easily see that $D_{H'}$ is a cycle isolating set of H' and $|D_{H'}| = t = \frac{|V(H')|}{4}$.

In all three situations, we conclude that $|D_{H'}| = \frac{|V(H')|}{4}$ and $x \in N[D_{H'}]$ (since $w \in D_{H'}$ and $x \in N(w)$).

Define $Y := V(H^*) \cup V(G_v)$ and $Z := V(H') \cup \{x\}$. Since H^* is linked to at least two vertices in $N(v)$, we observe that $E(V(H^*), N(v) \setminus \{x\}) \neq \emptyset$ and thus $G[Y]$ is connected. This shows that the components of $G - Z$ are $G[Y]$ and the members

of $\mathcal{H}_g^x \setminus \{H'\}$. Moreover, $G[Y] \not\cong C_3$ (because $|Y| = |V(H^*)| + |V(G_v)| \geq 4$).

Since $D_{H'}$ is a cycle isolating set of H' and $x \in N[D_{H'}]$, we see that $D_{H'}$ is also a cycle isolating set of $G[Z]$ and $E(Z \setminus N[D_{H'}], V(G) \setminus Z) = \emptyset$. By [Lemmas 2.1, 2.3](#) and [Theorem 1.3](#), we have

$$\begin{aligned} \iota_c(G) &\leq |D_{H'}| + \iota_c(G - Z) = \frac{|V(H')|}{4} + \iota_c(G[Y]) + \sum_{H \in \mathcal{H}_g^x \setminus \{H'\}} \iota_c(H) \\ &\leq \frac{|V(H')|}{4} + \frac{|Y|}{4} + \sum_{H \in \mathcal{H}_g^x \setminus \{H'\}} \frac{|V(H)|}{4} = \frac{|V(H')|}{4} + \frac{n-1-|V(H')|}{4} = \frac{n}{4} - \frac{1}{4} < \frac{n}{4}, \end{aligned}$$

a contradiction. ■

By [Claim 4](#), we know that $V(G) = X \cup V(G_v)$. Since $n \geq 12$ and $|X| = 4$, we have $|V(G_v)| \geq 8$ and hence $G_v \in \mathcal{G}_1 \cup \mathcal{G}_2$. There are two subcases to consider.

Subcase 2.2.1.1. $G_v \in \mathcal{G}_1$.

In this subcase, $G_v \cong A_i$ for some $i \in [10]$ (where the vertices of G_v are labeled as shown in [Fig. 2](#)) and $v = a_j$ for some $j \in [8]$. Moreover, $n = |X| + |V(G_v)| = 4 + 8 = 12$, and thus $\iota_c(G) = \frac{n}{4} = 3$.

Claim 5. For any $k \in [3]$ and any vertex $x' \in N(v) \setminus \{x\}$, if $y_k x \in E(G)$, then $y_k x' \notin E(G)$.

Proof. Suppose to the contrary that there exist some $k \in [3]$ and some vertex $x' \in N(v) \setminus \{x\}$ such that $y_k x, y_k x' \in E(G)$. Define $Y := X \cup \{x'\}$. It is clear that $\{y_k\}$ is a dominating set of $G[Y]$. Note that $G - Y = G_v - x'$. Then by [Observation 2.4\(i\)](#), $G - Y$ is connected. Since $|V(G - Y)| = 7$, we further have $G - Y \not\cong C_3$. By [Corollary 2.2](#) and [Theorem 1.3](#), we see that

$$\iota_c(G) \leq |\{y_k\}| + \iota_c(G - Y) \leq 1 + \frac{|V(G - Y)|}{4} = 1 + \frac{7}{4} = \frac{11}{4} < 3,$$

contradicting the assumption that $\iota_c(G) = 3$. ■

Suppose there exists some $k \in [8]$ such that a_k has degree 4 in G_v and $v \notin N_{G_v}[a_k]$. Define $Z := N_{G_v}[a_k]$. Then $|Z| = 5$ and $G_v - Z \cong C_3$; otherwise, $\{a_k\}$ is a cycle isolating set of G_v , contradicting [Lemma 2.5](#). Since $V(G - Z) = X \cup V(G_v - Z)$ and $xv \in E(G)$ with $x \in X$ and $v \in V(G_v - Z)$, we know that $G - Z$ is connected. Moreover, $G - Z \not\cong C_3$ (because $|V(G - Z)| = 7$). Since $\{a_k\}$ is a dominating set of $G[Z]$, it follows from [Corollary 2.2](#) and [Theorem 1.3](#) that

$$\iota_c(G) \leq |\{a_k\}| + \iota_c(G - Z) \leq 1 + \frac{|V(G - Z)|}{4} = 1 + \frac{7}{4} = \frac{11}{4} < 3,$$

a contradiction.

Therefore, we can conclude that such an a_k does not exist. Then by [Observation 2.4\(iii\)](#), one of the following holds:

- (a) $G_v \cong A_2$ and $v \in \{a_1, a_5\}$;
- (b) $G_v \cong A_6$ and $v \in \{a_5, a_6, a_7, a_8\}$;
- (c) $G_v \cong A_{10}$ and $v \in \{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8\}$.

- Suppose (a) holds. By symmetry, we may assume that $v = a_1$. Since v has degree 3 in G_v and $xv \in E(G)$, $\Delta(G) = d(v) = 4$. Note that $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\} = \{a_2, a_5, a_7\}$. Because $\Delta(G) = 4$, we notice that $|E(\{y_2, y_3\}, \{a_2, a_5\})| \leq 1$. Then, it is easy to check that $\{x, a_6\}$ is a cycle isolating set of G , which contradicts the assumption that $\iota_c(G) = 3$.
- Suppose (b) holds. By symmetry, we may assume that $v \in \{a_5, a_6\}$.
First, suppose $v = a_5$. Since v has degree 4 in G_v and $xv \in E(G)$, $\Delta(G) = d(v) = 5$. Moreover, we also have $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\} = \{a_1, a_6, a_7, a_8\}$. Because $y_1 x \in E(G)$, it follows from [Claim 5](#) that $E(\{y_1\}, \{a_1, a_6, a_7, a_8\}) = \emptyset$. Since H^* is linked to at least two vertices in $N(v)$, we may assume by symmetry that $E(\{y_2\}, \{a_1, a_6, a_7, a_8\}) \neq \emptyset$. This means that $y_3 x \in E(G)$; otherwise, suppose that $y_2 a_k \in E(G)$ for some $k \in \{1, 6, 7, 8\}$, then $\{a_k, a_{9-k}\}$ is a cycle isolating set of G , giving a contradiction. If $|E(\{y_2\}, \{a_6, a_7\})| \leq 1$, then $\{x, a_4\}$ is a cycle isolating set of G , a contradiction. Hence, $|E(\{y_2\}, \{a_6, a_7\})| = 2$ (i.e. $y_2 a_6, y_2 a_7 \in E(G)$). But then, it is straightforward to verify that either $\{y_2, a_4\}$ (if $xa_1 \notin E(G)$) or $\{x, a_8\}$ (if $xa_1 \in E(G)$) is a cycle isolating set of G , a contradiction.
Next, suppose $v = a_6$. Since v has degree 3 in G_v and $xv \in E(G)$, $\Delta(G) = d(v) = 4$. Note that $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\} = \{a_5, a_7, a_8\}$. Because $\Delta(G) = 4$, we conclude that $|E(\{y_2, y_3\}, \{a_7, a_8\})| \leq 1$. Then, one can easily see that $\{x, a_1\}$ is a cycle isolating set of G , a contradiction.
- Suppose (c) holds. By symmetry, we may assume that $v \in \{a_1, a_2\}$. Since G_v is cubic and $xv \in E(G)$, we have $\Delta(G) = d(v) = 4$.
First, suppose $v = a_1$. Then $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\} = \{a_2, a_3, a_5\}$. Since $\Delta(G) = 4$, each vertex in $\{a_2, a_3, a_5\}$ has at most one neighbor in $\{y_1, y_2, y_3\}$. This implies that $|E(\{y_2, y_3\}, \{a_2, a_3\})| \leq 2$. If $|E(\{y_2, y_3\}, \{a_2, a_3\})| \leq 1$, then $\{x, a_8\}$ is a cycle isolating set of G , a contradiction. Thus, $|E(\{y_2, y_3\}, \{a_2, a_3\})| = 2$. By symmetry, we may assume that $y_2 a_2 \in E(G)$. This shows that $y_3 x \in E(G)$; otherwise, $\{a_2, a_7\}$ is a cycle isolating set of G , a contradiction. Then,

it follows from Claim 5 that $y_3a_3 \notin E(G)$. Since $|E(\{y_2, y_3\}, \{a_2, a_3\})| = 2$, we know that $y_2a_3 \in E(G)$. But now, it is easy to verify that $\{y_2, a_6\}$ is a cycle isolating set of G , again a contradiction.

Next, suppose $v = a_2$. Then $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\} = \{a_1, a_3, a_4\}$. Since $\Delta(G) = 4$, each vertex in $\{a_1, a_3, a_4\}$ has at most one neighbor in $\{y_1, y_2, y_3\}$. This means that $|E(\{y_2, y_3\}, \{a_3, a_4\})| \leq 2$. If $|E(\{y_2, y_3\}, \{a_3, a_4\})| \leq 1$, then $\{x, a_5\}$ is a cycle isolating set of G , a contradiction. Therefore, $|E(\{y_2, y_3\}, \{a_3, a_4\})| = 2$. Without loss of generality, we may assume that $y_2a_3 \in E(G)$. If $y_3x \notin E(G)$, then $\{a_3, a_6\}$ is a cycle isolating set of G , a contradiction. Hence, we further conclude that $y_3x \in E(G)$. Then, it follows from Claim 5 that $y_3a_4 \notin E(G)$. Since $|E(\{y_2, y_3\}, \{a_3, a_4\})| = 2$, we have $y_2a_4 \in E(G)$. But then, it is easy to check that $\{y_2, a_5\}$ is a cycle isolating set of G , giving a contradiction.

Subcase 2.2.1.2. $G_v \in \mathcal{G}_2$.

Let $G_1, G_2, \dots, G_t$ be the C_3 -constituents of G_v , and for each $i \in [t]$, let u_i be the C_3 -connection of G_i and $H_i := u_i^1 u_i^2 u_i^3 u_i^4$ be the copy of C_3 in G_i . Since $|V(G_v)| \geq 8$, we see that $t \geq 2$. Recall that $G_i \in \{C_3^+, K_4^-, K_4\}$ for each $i \in [t]$.

Define $D_1 := \{u_i^1 : 1 \leq i \leq t\}$ and $D_2 := \{u_i : 1 \leq i \leq t\}$. Then, it is straightforward to verify that both D_1 and D_2 are cycle isolating sets of G_v and

$$|D_1| = |D_2| = \frac{n-4}{4} = \frac{n}{4} - 1 = \iota_c(G) - 1.$$

Moreover, D_1 is a dominating set of G_v .

Claim 6. For any $k \in [3]$, if $y_kx \in E(G)$, then $E(\{y_k\}, V(G_v) \setminus D_2) = \emptyset$.

Proof. Suppose to the contrary that there exists some $k \in [3]$ such that $y_kx \in E(G)$ and $E(\{y_k\}, V(G_v) \setminus D_2) \neq \emptyset$. Then $y_k u_i^j \in E(G)$ for some $i \in [t]$ and $j \in [3]$. It is easy to observe that $(D_2 \setminus \{u_i\}) \cup \{y_k\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. ■

Claim 7. If $E(V(H^*), D_2) \neq \emptyset$, then $v \in V(G_v) \setminus D_2$.

Proof. Suppose to the contrary that $v \in D_2$. Since $xv \in E(G)$ and $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\}$, we derive that $x \in N[D_2]$ and $N(V(H^*)) \cap V(G_v) \subseteq N[D_2]$. Moreover, because $E(V(H^*), D_2) \neq \emptyset$, there exists some $k \in [3]$ such that $y_k \in N[D_2]$. This implies that D_2 is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. ■

Note that $G[X] \in \{C_3^+, K_4^-, K_4\}$. We now consider all three possibilities of $G[X]$.

- Suppose $G[X] \cong C_3^+$. Then $y_1x \in E(G)$ and $y_2x, y_3x \notin E(G)$. By Claim 6, $E(\{y_1\}, V(G_v) \setminus D_2) = \emptyset$.

First, suppose $E(\{y_1\}, D_2) \neq \emptyset$. Then there exists some $i \in [t]$ such that $y_1 u_i^j \in E(G)$. Since $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\}$ and by Claim 7, we can conclude that $v = u_i^j$ for some $j \in [3]$ with $u_i u_i^j \in E(G)$. This shows that $\Delta(G) = d(v) = 4$ (because v has degree 3 in G_v and $xv \in E(G)$). Notice that $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\} = \{u_i, u_i^k, u_i^s\}$ (where $k, s \in [3] \setminus \{j\}$). If $|E(\{y_2, y_3\}, \{u_i^k, u_i^s\})| \leq 1$, then $(D_2 \setminus \{u_i\}) \cup \{x\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. Thus, $|E(\{y_2, y_3\}, \{u_i^k, u_i^s\})| \geq 2$. Without loss of generality, we may assume that $y_2 u_i^k \in E(G)$. If $y_2 u_i^s \in E(G)$, then it is easy to see either $(D_1 \setminus \{u_i^1\}) \cup \{y_2\}$ (if $xu_i \notin E(G)$) or $(D_2 \setminus \{u_i\}) \cup \{y_2\}$ (if $xu_i \in E(G)$) is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. Similarly, if $y_3 u_i^k \in E(G)$, then either $(D_1 \setminus \{u_i^1\}) \cup \{u_i^k\}$ (if $xu_i \notin E(G)$) or $(D_2 \setminus \{u_i\}) \cup \{u_i^k\}$ (if $xu_i \in E(G)$) is a cycle isolating set of G of size $\iota_c(G) - 1$, again a contradiction. Therefore, $y_2 u_i^s, y_3 u_i^k \notin E(G)$. Since $|E(\{y_2, y_3\}, \{u_i^k, u_i^s\})| \geq 2$, we derive that $y_3 u_i^s \in E(G)$. If $xu_i \in E(G)$, then $(D_2 \setminus \{u_i\}) \cup \{u_i^k\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. If $y_3 u_i \in E(G)$, then one can easily check that either $(D_1 \setminus \{u_i^1\}) \cup \{y_3\}$ (if $xu_i^k \notin E(G)$) or $(D_2 \setminus \{u_i\}) \cup \{u_i^k\}$ (if $xu_i^k \in E(G)$) is a cycle isolating set of G of size $\iota_c(G) - 1$, again a contradiction. Hence, we further have $xu_i, y_3 u_i \notin E(G)$. But then, $(D_1 \setminus \{u_i^1\}) \cup \{u_i^k\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, giving a contradiction.

Next, suppose $E(\{y_1\}, D_2) = \emptyset$ and $E(\{y_2, y_3\}, D_2) \neq \emptyset$. By symmetry, we may assume that $E(\{y_2\}, D_2) \neq \emptyset$. Then there exists some $i \in [t]$ such that $y_2 u_i^j \in E(G)$, and thus it follows from Claim 7 and $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\}$ that $v = u_i^j$ for some $j \in [3]$ with $u_i u_i^j \in E(G)$. Since v has degree 3 in G_v and $xv \in E(G)$, $\Delta(G) = d(v) = 4$. Then $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\} = \{u_i, u_i^k, u_i^s\}$ (where $k, s \in [3] \setminus \{j\}$). Moreover, because $E(\{y_1\}, D_2) = \emptyset$, we see that $y_1 u_i \notin E(G)$. If $|E(\{y_2, y_3\}, \{u_i^k, u_i^s\})| \leq 1$, then $(D_2 \setminus \{u_i\}) \cup \{x\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. Therefore, $|E(\{y_2, y_3\}, \{u_i^k, u_i^s\})| \geq 2$. Note that $|E(\{y_2\}, \{u_i^k, u_i^s\})| \leq 1$ (since $\Delta(G) = 4$). Hence, we have $|E(\{y_3\}, \{u_i^k, u_i^s\})| \geq 1$. Without loss of generality, we may assume that $y_3 u_i^k \in E(G)$. If $y_3 u_i^s \in E(G)$, then either $(D_1 \setminus \{u_i^1\}) \cup \{y_3\}$ (if $xu_i \notin E(G)$) or $(D_2 \setminus \{u_i\}) \cup \{y_3\}$ (if $xu_i \in E(G)$) is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. If $y_2 u_i^k \in E(G)$, then $(D_1 \setminus \{u_i^1\}) \cup \{u_i^k\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, again a contradiction. Thus, $y_3 u_i^s, y_2 u_i^k \notin E(G)$. Since $|E(\{y_2, y_3\}, \{u_i^k, u_i^s\})| \geq 2$, we can derive that $y_2 u_i^s \in E(G)$. Then, it is easy to verify that either $(D_1 \setminus \{u_i^1\}) \cup \{y_2\}$ (if $xu_i^k \notin E(G)$) or $(D_2 \setminus \{u_i\}) \cup \{u_i^k\}$ (if $xu_i^k \in E(G)$) is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction.

Finally, suppose $E(V(H^*), D_2) = \emptyset$. If there exist some $i \in [t]$ and $j, k \in [3]$ such that $xu_i^j, xu_i^k \in E(G)$, then we let $s \in [3] \setminus \{j, k\}$ and it is easy to check that either $(D_1 \setminus \{u_i^1\}) \cup \{x\}$ (if $|E(\{y_2, y_3\}, \{u_i^s\})| \leq 1$) or $(D_1 \setminus \{u_i^1\}) \cup \{u_i^s\}$ (if

$|E(\{y_2, y_3\}, \{u_i^s\})| = 2)$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. Therefore, $|E(\{x\}, \{u_i^1, u_i^2, u_i^3\})| \leq 1$ for each $i \in [t]$. Since H^* is linked to at least two vertices in $N(v)$, we observe that $E(V(H^*), N(v) \setminus \{x\}) \neq \emptyset$ and hence $E(V(H^*), V(G_v)) \neq \emptyset$. Then, it follows from $E(V(H^*), D_2) = \emptyset$ and $E(\{y_1\}, V(G_v) \setminus D_2) = \emptyset$ that $E(\{y_2, y_3\}, V(G_v) \setminus D_2) \neq \emptyset$. By symmetry, we may assume that $E(\{y_2\}, V(G_v) \setminus D_2) \neq \emptyset$. Then there exist some $i \in [t]$ and $j \in [3]$ such that $y_2 u_i^j \in E(G)$. But now, $(D_2 \setminus \{u_i\}) \cup \{y_2\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, giving a contradiction.

- Suppose $G[X] \cong K_4^-$. Without loss of generality, we may assume that $y_1 x, y_2 x \in E(G)$ and $y_3 x \notin E(G)$. Then by Claim 6, $E(\{y_1, y_2\}, V(G_v) \setminus D_2) = \emptyset$.

First, suppose $E(V(H^*), D_2) \neq \emptyset$. Then there exist some $p \in [3]$ and $i \in [t]$ such that $y_p u_i \in E(G)$. Since $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\}$ and by Claim 7, we see that $v = u_i^j$ for some $j \in [3]$ with $u_i u_i^j \in E(G)$. This implies that $\Delta(G) = d(v) = 4$ (because v has degree 3 in G_v and $xv \in E(G)$). Note that $N(V(H^*)) \cap V(G_v) \subseteq N(v) \setminus \{x\} = \{u_i, u_i^k, u_i^s\}$ (where $k, s \in [3] \setminus \{j\}$). If $p \in [2]$, then either $(D_2 \setminus \{u_i\}) \cup \{x\}$ (if $|E(\{y_3\}, \{u_i^k, u_i^s\})| \leq 1$) or $(D_2 \setminus \{u_i\}) \cup \{y_3\}$ (if $|E(\{y_3\}, \{u_i^k, u_i^s\})| = 2$) is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. Thus, $p = 3$. Since $\Delta(G) = 4$, we notice that $|E(\{y_3\}, \{u_i^k, u_i^s\})| \leq 1$. Then, it is easy to see that $(D_2 \setminus \{u_i\}) \cup \{x\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, again a contradiction.

Next, suppose $E(V(H^*), D_2) = \emptyset$. If there exist some $i \in [t]$ and $j, k \in [3]$ such that $x u_i^j, x u_i^k \in E(G)$, then $(D_1 \setminus \{u_i^1\}) \cup \{x\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction. Hence, $|E(\{x\}, \{u_i^1, u_i^2, u_i^3\})| \leq 1$ for each $i \in [t]$. Since H^* is linked to at least two vertices in $N(v)$, we know that $E(V(H^*), N(v) \setminus \{x\}) \neq \emptyset$ and thus $E(V(H^*), V(G_v)) \neq \emptyset$. Then, it follows from $E(V(H^*), D_2) = \emptyset$ and $E(\{y_1, y_2\}, V(G_v) \setminus D_2) = \emptyset$ that $E(\{y_3\}, V(G_v) \setminus D_2) \neq \emptyset$. This means that there exist some $i \in [t]$ and $j \in [3]$ such that $y_3 u_i^j \in E(G)$. But then, $(D_2 \setminus \{u_i\}) \cup \{y_3\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, a contradiction.

- Suppose $G[X] \cong K_4$. Then $y_1 x, y_2 x, y_3 x \in E(G)$, which shows that $E(V(H^*), V(G_v) \setminus D_2) = \emptyset$ (by Claim 6). On the other hand, since H^* is linked to at least two vertices in $N(v)$, we conclude that $E(V(H^*), N(v) \setminus \{x\}) \neq \emptyset$ and hence $E(V(H^*), V(G_v)) \neq \emptyset$. Therefore, $E(V(H^*), D_2) \neq \emptyset$. Then by Claim 7, we see that $v = u_i^j$ for some $i \in [t]$ and $j \in [3]$. But now, since $y_1 x, y_2 x, y_3 x, xv \in E(G)$, one can easily verify that $(D_2 \setminus \{u_i\}) \cup \{x\}$ is a cycle isolating set of G of size $\iota_c(G) - 1$, giving a contradiction.

Subcase 2.2.2. $G_v \cong C_3$.

In this subcase, $\Delta(G) = d(v) = 3$ (since v has degree 2 in G_v and $xv \in E(G)$). Define $Y := X \cup V(G_v)$. Then $|Y| = 7$ and the components of $G - Y$ are the members of $\mathcal{H}_g^x$. Since $n \geq 12$ and $|Y| = 7$, we have $V(G) \setminus Y \neq \emptyset$ (i.e. $\mathcal{H}_g^x \neq \emptyset$). Moreover, because G is connected and $\Delta(G) = 3$, $\mathcal{H}_g^x$ has exactly one member H and $|E(\{x\}, V(H))| = 1$.

Since $xv \in E(G)$ and $|Y| = 7$, we deduce that $G[Y]$ is connected and $G[Y] \not\cong C_3$. Then by Theorem 1.3, $\iota_c(G[Y]) \leq \frac{|Y|}{4} = \frac{7}{4}$, i.e. $\iota_c(G[Y]) = 1$. Let D be a cycle isolating set of $G[Y]$ with $|D| = 1$. Since $|E(\{x\}, V(H))| = 1$, each component of $G[Y] - N[D]$ joins to $V(G) \setminus Y$ by at most one edge. Then by Lemma 2.1 and Theorem 1.3, we have

$$\iota_c(G) \leq |D| + \iota_c(G - Y) = 1 + \iota_c(H) \leq 1 + \frac{|V(H)|}{4} = 1 + \frac{n-7}{4} = \frac{n}{4} - \frac{3}{4} < \frac{n}{4},$$

a contradiction.

This completes the proof of Theorem 1.4. ■

4. Proof of Theorem 1.5

In this section, we give the proof of Theorem 1.5.

Proof of Theorem 1.5. Let G be a connected n -vertex graph with $\iota(G, K_3) = \frac{n}{4}$. It is clear that for any $D \subseteq V(G)$, if $G - N[D]$ contains no cycle, then $G - N[D]$ contains no K_3 . This implies that any cycle isolating set of G must also be a K_3 -isolating set of G , and thus $\iota(G, K_3) \leq \iota_c(G)$. Then by Theorem 1.3, we see that

$$\iota(G, K_3) \leq \iota_c(G) \leq \frac{n}{4}. \quad (5)$$

Since $\iota(G, K_3) = \frac{n}{4}$, we derive that all inequalities in (5) must be equalities, which means that $\iota_c(G) = \frac{n}{4}$. Then, it follows from Theorem 1.4 that $G \in \{C_4\} \cup \mathcal{G}_1 \cup \mathcal{G}_2$. If $G \cong C_4$, then $n = 4$ and $\iota(G, K_3) = 0 < 1 = \frac{n}{4}$, a contradiction. If $G \cong A_2$, then $n = 8$ and it is straightforward to check that $\{a_1\}$ is a K_3 -isolating set of G , which shows that $\iota(G, K_3) = 1 < 2 = \frac{n}{4}$, again a contradiction. Therefore, we further deduce that $G \in (\mathcal{G}_1 \setminus \{A_2\}) \cup \mathcal{G}_2$.

Now, in order to prove Theorem 1.5, it suffices to show that $\iota(G, K_3) = \frac{n}{4}$ for every n -vertex graph $G \in (\mathcal{G}_1 \setminus \{A_2\}) \cup \mathcal{G}_2$.

First, suppose $G \in \mathcal{G}_1 \setminus \{A_2\}$. Then $n = 8$ and $G \cong A_i$ for some $i \in [10] \setminus \{2\}$. It is easy to verify that for any $j \in [8]$, $G - N[a_j]$ contains a K_3 . This implies that $\iota(G, K_3) \geq 2$. On the other hand, by Theorem 1.1, we know that $\iota(G, K_3) \leq \frac{n}{4} = 2$. Hence, $\iota(G, K_3) = 2 = \frac{n}{4}$.

Next, suppose $G \in \mathcal{G}_2$. Let $G_1, G_2, \dots, G_t$ be the C_3 -constituents of G , and for each $i \in [t]$, let u_i be the C_3 -connection of G_i and $H_i := u_i^1 u_i^2 u_i^3 u_i^1$ be the copy of C_3 in G_i . Then $n = 4t$. One can easily check that for any K_3 -isolating set D of G , we must have $D \cap V(G_i) \neq \emptyset$ for each $i \in [t]$; otherwise, some H_i would be a K_3 in $G - N[D]$, giving a contradiction. This shows that $\iota(G, K_3) \geq t$. On the other hand, it follows from Theorem 1.1 that $\iota(G, K_3) \leq \frac{n}{4} = t$. Thus, $\iota(G, K_3) = t = \frac{n}{4}$.

This completes the proof of Theorem 1.5. ■

Acknowledgments

The authors would like to thank the anonymous referees for their careful reading and valuable suggestions which have improved the presentation of this paper. This work was partially supported by the National Natural Science Foundation of China (Nos. 12171239 and 12271251).

Data availability

No data was used for the research described in the article.

References

- [1] K. Bartolo, P. Borg, D. Scicluna, Isolation of squares in graphs, *Discrete Math.* 347 (2024) 114161.
- [2] P. Borg, Isolation of cycles, *Graphs Combin.* 36 (2020) 631–637.
- [3] P. Borg, Isolation of connected graphs, *Discrete Appl. Math.* 339 (2023) 154–165.
- [4] P. Borg, Isolation of regular graphs, stars and k -chromatic graphs, 2023, arXiv preprint, arXiv:2303.13709.
- [5] P. Borg, Isolation of regular graphs and k -chromatic graphs, *Mediterr. J. Math.* 21 (2024) 148.
- [6] P. Borg, K. Fenech, P. Kaemawichanurat, Isolation of k -cliques, *Discrete Math.* 343 (2020) 111879.
- [7] P. Borg, K. Fenech, P. Kaemawichanurat, Isolation of k -cliques II, *Discrete Math.* 345 (2022) 112641.
- [8] P. Borg, P. Kaemawichanurat, Partial domination of maximal outerplanar graphs, *Discrete Appl. Math.* 283 (2020) 306–314.
- [9] P. Borg, P. Kaemawichanurat, Extensions of the art gallery theorem, *Ann. Comb.* 27 (2023) 31–50.
- [10] G. Boyer, W. Goddard, Disjoint isolating sets and graphs with maximum isolation number, *Discrete Appl. Math.* 356 (2024) 110–116.
- [11] G. Boyer, W. Goddard, M.A. Henning, On total isolation in graphs, *Aequat. Math.* (2024) <http://dx.doi.org/10.1007/s00010-024-01057-1>.
- [12] Y. Caro, A. Hansberg, Partial domination - the isolation number of a graph, *Filomat* 31 (2017) 3925–3944.
- [13] J. Chen, S.-J. Xu, P_5 -Isolation in graphs, *Discrete Appl. Math.* 340 (2023) 331–349.
- [14] Q. Cui, J. Zhang, A sharp upper bound on the cycle isolation number of graphs, *Graphs Combin.* 39 (2023) 117.
- [15] O. Favaron, P. Kaemawichanurat, Inequalities between the K_k -isolation number and the independent K_k -isolation number of a graph, *Discrete Appl. Math.* 289 (2021) 93–97.
- [16] J.F. Fink, M.S. Jacobson, L.F. Kinch, J. Roberts, On graphs having domination number half their order, *Period. Math. Hungar.* 16 (1985) 287–293.
- [17] W. Goddard, M.A. Henning, Independent domination in graphs: A survey and recent results, *Discrete Math.* 313 (2013) 839–854.
- [18] T.W. Haynes, S.T. Hedetniemi, M.A. Henning (Eds.), *Topics in Domination in Graphs*, in: *Developments in Mathematics*, vol. 64, Springer, Cham, 2020.
- [19] T.W. Haynes, S.T. Hedetniemi, M.A. Henning (Eds.), *Structures of Domination in Graphs*, in: *Developments in Mathematics*, vol. 66, Springer, Cham, 2021.
- [20] T.W. Haynes, S.T. Hedetniemi, M.A. Henning, *Domination in Graphs: Core Concepts*, in: *Springer Monographs in Mathematics*, Springer, Cham, 2023.
- [21] T.W. Haynes, S.T. Hedetniemi, P.J. Slater, *Fundamentals of Domination in Graphs*, Marcel Dekker Inc., New York, 1998.
- [22] T.W. Haynes, S.T. Hedetniemi, P.J. Slater (Eds.), *Domination in Graphs: Advanced Topics*, Marcel Dekker Inc., New York, 1998.
- [23] M.A. Henning, A survey of selected recent results on total domination in graphs, *Discrete Math.* 309 (2009) 32–63.
- [24] M.A. Henning, A. Yeo, *Total Domination in Graphs*, in: *Springer Monographs in Mathematics*, Springer, New York, 2013.
- [25] H. Hua, X. Hua, Admissible property of graphs in terms of independence number, *Bull. Malays. Math. Sci. Soc.* 45 (2022) 2123–2135.
- [26] Y. Huang, G. Zhang, X. Jin, New results on the 1-isolation number of graphs without short cycles, 2023, arXiv preprint, arXiv:2308.00581.
- [27] P. Kaemawichanurat, O. Favaron, Partial domination and irredundance numbers in graphs, *Appl. Math. Comput.* 457 (2023) 128153.
- [28] M. Lemańska, M. Mora, M.J. Souto-Salorio, Graphs with isolation number equal to one third of the order, *Discrete Math.* 347 (2024) 113903.
- [29] O. Ore, *Theory of Graphs*, American Mathematical Society Colloquium Publications, vol. 38, American Mathematical Society, Providence, Rhode Island, 1962.
- [30] C. Payan, N.H. Xuong, Domination-balanced graphs, *J. Graph Theory* 6 (1982) 23–32.
- [31] S. Tokunaga, T. Jiarasuksakun, P. Kaemawichanurat, Isolation number of maximal outerplanar graphs, *Discrete Appl. Math.* 267 (2019) 215–218.
- [32] X. Wei, G. Zhang, B. Zhao, On the C_4 -isolation number of a graph, 2023, arXiv preprint, arXiv:2310.17337.
- [33] J. Yan, Isolation of the diamond graph, *Bull. Malays. Math. Sci. Soc.* 45 (2022) 1169–1181.
- [34] Y. Yin, X. An, B. Wu, $K_{1,2}$ -Isolation number of claw-free cubic graphs, *Bull. Malays. Math. Sci. Soc.* 47 (2024) 75.
- [35] H. Yu, B. Wu, Admissible property of graphs in terms of radius, *Graphs Combin.* 38 (2022) 6.
- [36] G. Zhang, B. Wu, $K_{1,2}$ -Isolation in graphs, *Discrete Appl. Math.* 304 (2021) 365–374.
- [37] G. Zhang, B. Wu, Isolation of cycles and trees in graphs, *J. Xinjiang Univ. (Nat. Sci.)* 39 (2022) 169–175.
- [38] G. Zhang, B. Wu, Cycle isolation of graphs with small girth, *Graphs Combin.* 40 (2024) 38.
- [39] G. Zhang, B. Wu, k -Isolation in graphs, *Discrete Appl. Math.* 357 (2024) 99–111.
- [40] G. Zhang, B. Wu, A note on the cycle isolation number of graphs, *Bull. Malays. Math. Sci. Soc.* 47 (2024) 57.
- [41] G. Zhang, B. Wu, On the cycle isolation number of triangle-free graphs, *Discrete Math.* 347 (2024) 114190.